STATEBOOK

A Coherence Layer for Frontier-Risk Markets

A coherence layer for frontier-risk markets, with explicit semantic, payoff, execution, capital, settlement, assurance, and recovery boundaries.

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No execution, custody, pause, signing, margin-recognition, oracle, routing, bridge, or settlement authority. No production-readiness, legal, full-security, or final-incident-attribution claim.

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A Coherence Layer for Frontier-Risk Markets

Assurance-adjusted externalization, capital completeness, and financial infrastructure for the AI economy

State slice: `statebook-whitepaper-prd-and-publication-media-boundary`.

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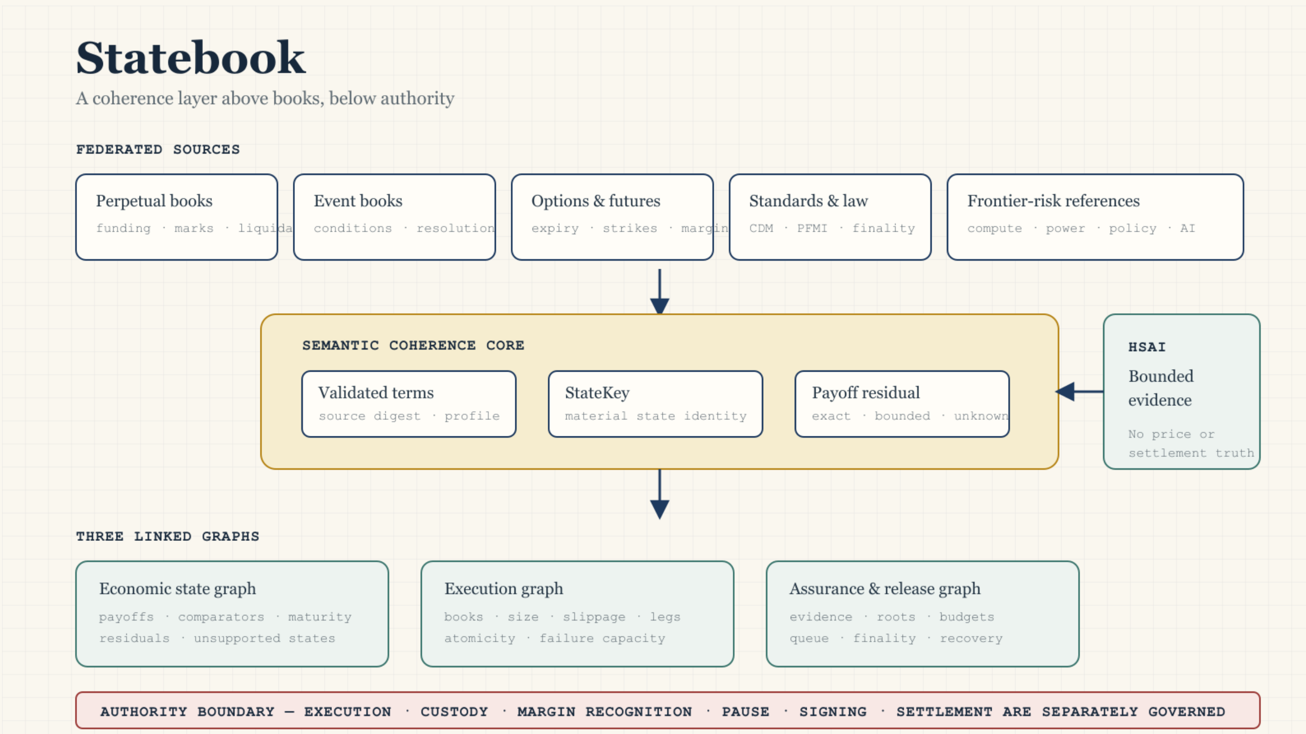
This paper is architecture research. It is not investment advice, legal advice, a product launch, production evidence, incident attribution, or authority to execute trades, recognize margin, hold collateral, pause a venue, sign a transaction, or settle value.



Statebook layer map · explanatory media, not evidence

Abstract

Financial venues are converging in product surface while remaining fragmented in semantics, execution, collateral, law, and settlement. Perpetual futures, event contracts, options, tokenized claims, compute futures, energy contracts, and other synthetic exposures increasingly describe different pieces of the same economic world. An order book can match orders in one instrument. It cannot determine whether two instruments observe the same state, whether a portfolio replicates a target payoff, whether all required legs can be filled, whether a clearing regime recognizes the offset, or whether value can settle with compatible legal finality.

This paper defines a Statebook as a semantic and risk-coherence layer above federated source books. The Statebook normalizes source terms into a bounded contract representation, groups contracts by precisely defined paying states,

computes state-by-state portfolio residuals, and reports seven independent dimensions: semantic, payoff, execution, capital, settlement, assurance, and recovery completeness. It preserves perpetuals as path-dependent hedge profiles rather than forcing them into terminal-claim semantics.

The paper also rejects a false binary between trust and instant settlement. Atomic linked settlement can reduce principal risk. Immediate irreversible externalization can magnify loss when an oracle, signer, calculation, bridge, or governance path is compromised. The proposed controller therefore leaves valid internal risk updates fast, permits all-or-none linked exchange only under strict conditions, and applies evidence-specific gates, loss budgets, and challenge windows to unilateral value release. Hard gates decide whether a release is eligible; multi-axis budgets decide how much; the maximum of named risk clocks decides when; and a challengeable state machine decides what happens next. No scalar reputation score can compensate for failed current evidence.

AI accelerates both sides of the design. It creates new hedgeable inputs and outcomes — compute capacity, power, model performance, data access, chips, latency, regulation, and labour displacement — while increasing machine-speed search, persuasion, exploit replication, and correlated automation. The likely next decade is therefore not simply a larger derivatives market. It is a race to build a programmable market substrate whose semantic, capital, security, and legal controls can progress at different clocks without being conflated.

1. Executive thesis

The phrase “AWS of finance” is useful only when taken beyond marketing. Amazon Web Services made computing infrastructure programmable and shifted many users from owning dedicated capacity to consuming standardized, metered services [M00]. A financial analogue would expose reusable execution, collateral, risk, lifecycle, and settlement primitives so that a new market does not require rebuilding an exchange from zero. Hyperliquid's expanding market-deployment surface, Kalshi's perpetual-futures specification, Architect's compute-market work, tokenized securities, and institutional moves toward longer trading hours are current signals of this convergence [M01-M04, I10-I12]. Jeff Yan's “AWS of finance” framing is Fortune's attribution of a founder thesis, not independent validation [X01-X02].

The analogy breaks at the point where finance differs from ordinary cloud compute:

- Financial state is an entitlement and obligation graph, not merely data.
- A duplicated server request is inconvenient; a duplicated payout can be a permanent loss.
- Product semantics are governed by benchmark, legal, calendar, disruption, default, and settlement rules.
- Capital offsets depend on an authority's recognized model and enforceable netting set, not mathematical correlation alone.
- Settlement finality can be technical, economic, accounting, and legal at different times.
- The provider can become the concentrated default, governance, surveillance, oracle, and operational-risk domain.

The resulting thesis is narrower and stronger:

> *A Statebook is a coherence layer over federated books. It organizes instruments by state-dependent payoff, identifies bounded replication candidates, exposes residual risk, and reports whether execution, collateral, settlement, and assurance conditions actually support the proposed action.*

It is not a universal clearinghouse, a price oracle, a legal-equivalence engine, or an automatic proof that the market is complete.

2. Why the order book is insufficient

An order book answers a local question: at what prices and quantities are participants willing to exchange one identified contract? When instruments share an economic reference but differ in payoff form, maturity, settlement, or venue, the important questions move above the individual book [T11A].

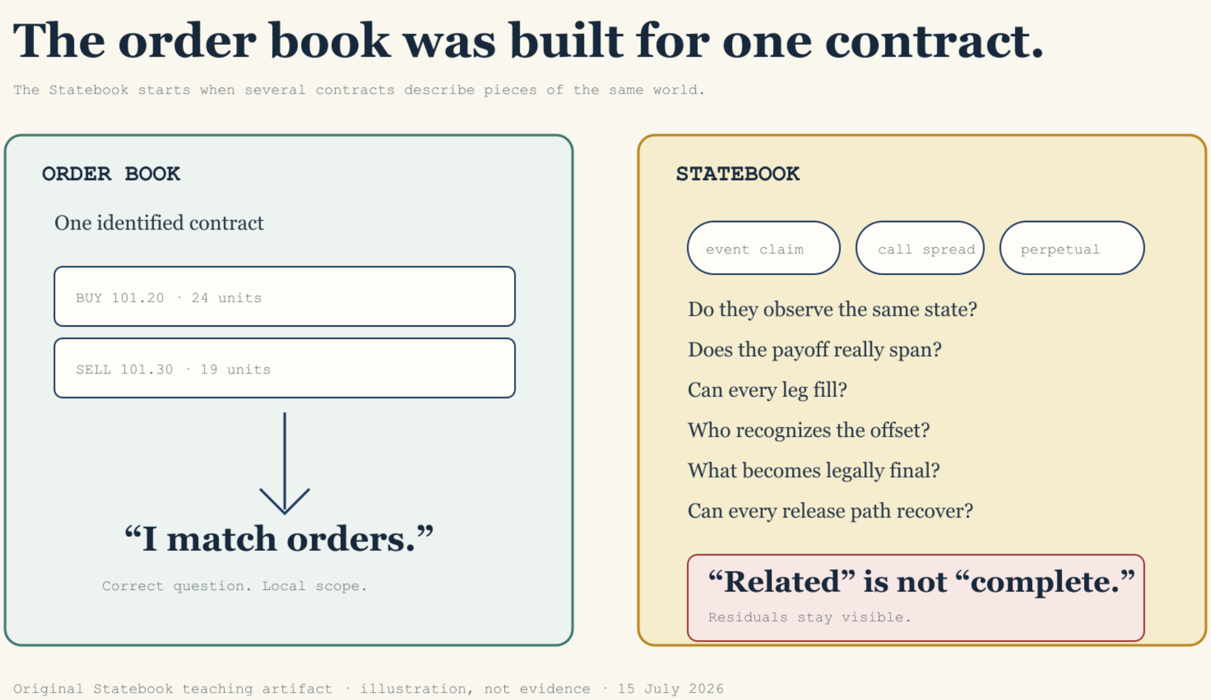
Consider three claims involving the same reference variable `X`:

1. a binary contract paying one unit if `X >= K` at time `T`;
2. a call spread spanning strikes around `K` and expiring at `T`;
3. a perpetual position whose mark, funding, collateral, and liquidation evolve continuously before and after `T`.

They may express related views. They are not the same contract. The binary may use a different observation window, benchmark administrator, correction rule, currency, payout cap, or dispute process. The option spread may approximate a step payoff only at finite strike width. The perpetual remains path dependent and can liquidate before the terminal observation occurs. Placing all three in one interface does not make them one coherent market.

The Statebook problem begins where the order book ends:

- What exact world state causes each claim to pay?
- Which terms are material to identity rather than display metadata?
- Is replication exact, bounded, approximate, or unsupported?
- Can all legs be executed at the claimed size and time?
- Is the offset recognized before liquidation?
- Are settlement rights, finality, and default rules compatible?
- Is the evidence authorizing irreversible release current and independent?



From order book to Statebook · explanatory media, not evidence

3. Intellectual lineage

3.1 State-contingent claims

Arrow and Debreu provide the foundational idea: economic goods and securities are indexed not only by physical description but by time and the state in which they are delivered [T01-T02]. Radner explains sequential trading without a complete date-zero claim set [T02B]. Hart shows that incomplete market structures can produce constrained inefficiency [T02C]. Duffie and Shafer establish generic existence of equilibrium in specified incomplete-market models [T02D].

Statebook adopts the indexing insight but rejects the frictionless conclusion. Its state identity includes reference methodology, observation interval, calendar, comparator, settlement asset, rounding, correction, dispute, default, and finality domain. Two tickers that look alike can represent distinct states. Two different product labels can sometimes represent related payoffs. Neither relation is inferred from a name.

3.2 Replication and state prices

Black-Scholes, Harrison-Pliska, Ross, and Breeden-Litzenberger establish the formal relationship among no arbitrage, replication, market completeness, and state-contingent prices [T03, T04, T05, T05A]. This lineage justifies a payoff algebra. It also supplies the most important warning: a mathematical replicating strategy depends on its model, trading set, admissible states, and execution assumptions.

Statebook therefore never equates “same reference” with “same exposure.” Exact replication is a quantified claim over a declared state domain. Approximate replication reports the residual and unsupported states. A historical correlation, model price, or point estimate cannot silently become exact.

3.3 Executable contract semantics

Compositional contract languages and ACTUS show that financial terms can be represented as typed event and cash-flow semantics rather than opaque prose. Peyton Jones, Eber, and Seward demonstrate compositional financial contracts; Annenkov and Elsman show verified compilation relative to formalized semantics; ACTUS defines standardized contract events and cash-flow schedules. FINOS CDM, FpML, ISO 20022, and the UPI ecosystem supply product, lifecycle, messaging, and reference-data interoperability [T05B, T05C, T05D, I13, I13A, I13B, I13C].

Statebook should compose with those standards where their semantics fit. It should not invent an all-purpose replacement ontology. Its differentiated job is cross-payoff state coherence, residual analysis, completeness typing, and evidence-bound action proposals.

3.4 Microstructure and information markets

Kyle and Glosten-Milgrom explain why execution cost, adverse selection, and price impact are endogenous [T10-T11]. Hanson demonstrates combinatorial information markets, and Wolfers-Zitzewitz synthesize prediction-market design [T13-T15]. Manski supplies a necessary caveat: a binary contract price is not automatically an objective probability [T13A]. It is a market-implied price whose probabilistic interpretation depends on beliefs, risk preferences, wealth, fees, constraints, and design.

The Statebook may expose implied distributions or bounds, but it must retain those assumptions. It cannot label every event price as a probability or every apparent arbitrage as executable profit.

3.5 Clearing and finality

The PFMI separate legal basis, credit, collateral, margin, liquidity, settlement, operational risk, links, and default management [I01]. Duffie-Zhu show that central clearing can either improve or destroy netting efficiency depending on product scope [I06]. SPAN and STANS demonstrate portfolio-level risk methods inside governed clearing universes [I08-I09]. CPMI work on DvP, PvP, fast payments, and tokenisation shows that atomicity and speed have several effects: linked exchange can reduce principal risk, while immediate gross settlement can raise prefunding and liquidity pressure [I02, I02A, I03, I04, I05].

The implication is decisive: payoff similarity alone cannot create capital or settlement completeness. Those dimensions require recognized authority, enforceable legal arrangements, operational capacity, and compatible finality.

4. Market convergence without coherence

4.1 Product forms are becoming primitives

“Prediction market,” “perpetual exchange,” and “options venue” increasingly describe payoff forms and distribution histories rather than permanent venue boundaries. Hyperliquid HIP-3 permits builder-deployed perpetual markets with custom oracles and settlement parameters [M02]. Kalshi publishes a BTC perpetual specification [M03]. Architect is developing compute-linked products [M04]. Gnosis Conditional Tokens and Polymarket's negative-risk adapter expose programmable outcome positions [M05-M07]. CoW Protocol demonstrates solver and batch-auction infrastructure [M08-M09].

This does not mean every venue will converge technically or legally. It means users will increasingly encounter the same underlying risk through several payoff forms and control domains.

4.2 Frontier-risk markets

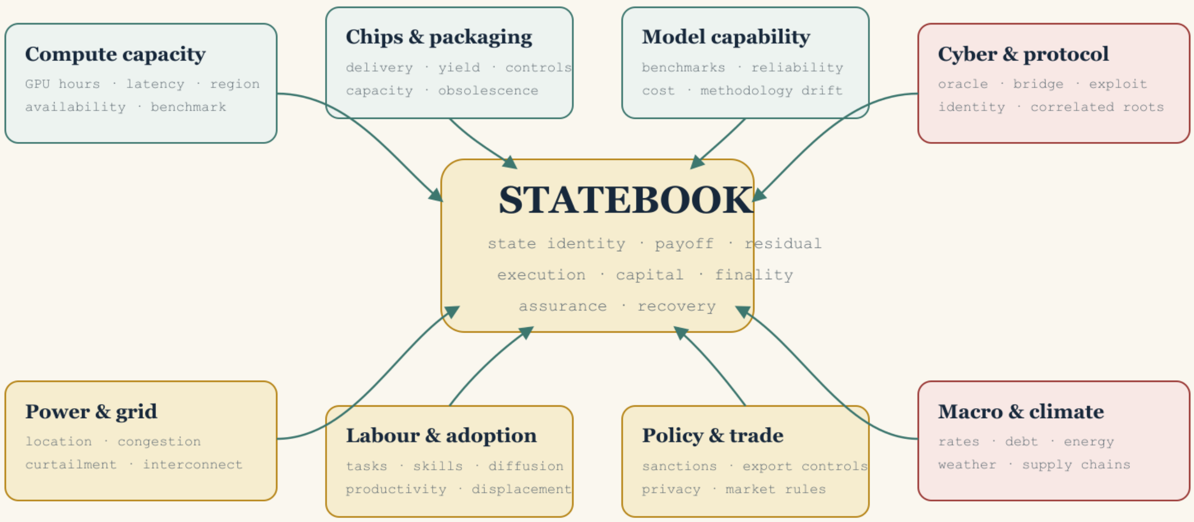
The term frontier-risk markets refers here to markets for rapidly emerging, hard-to-warehouse, or synthetically represented risks whose measurement and institutional boundaries are still forming. Examples include:

- GPU and accelerator-hour availability;
- data-centre power and congestion;
- grid interconnection and curtailment;
- chip delivery and advanced-packaging capacity;
- model inference price, latency, and service availability;
- benchmarked model capability or failure thresholds;
- data licensing and privacy-regulatory outcomes;
- cyber-loss, oracle, bridge, and protocol risk;
- carbon, weather, catastrophe, and adaptation outcomes;
- election, policy, sanctions, and trade-control events;
- autonomous-agent performance and liability triggers.

These underlyings are heterogeneous. A compute future may reference a benchmark rather than deliver a specific machine. Power is locational and non-storable. Model capability benchmarks can saturate or change methodology. Event contracts may face public-interest restrictions. Statebook's value is not to erase these differences. It is to make them machine-visible.

Frontier-risk markets

New payoff forms emerge faster than common semantics, infrastructure, and law.



Related underlyings do not imply shared books, margin, oracle governance, or legal finality.

Frontier-risk market map · explanatory media, not evidence

4.3 The structured-product implication

Once payoff forms become composable primitives, product design moves from listing a single contract to assembling state-contingent portfolios. A user could express “AI infrastructure boom without regional power-price shock,” “chip-delivery delay conditional on export restriction,” or “model-capability milestone with downside protected by compute-cost exposure.” This is structured- product territory.

Combinatorics increase both usefulness and danger. Every added leg introduces reference, basis, liquidity, default, legal, and settlement dependencies. A Statebook should make that residual graph explicit before it optimizes routing or margin.

5. The Statebook model

5.1 Bounded first domain

The first implementation domain should be intentionally narrow:

- terminal rather than perpetual;
- scalar rather than multidimensional;
- cash settled rather than physical;
- synthetic fixture data rather than live venues;

- exact rational arithmetic rather than floating point;
- deterministic local analysis rather than authority-bearing execution.

This scope is not lack of ambition. It is the minimum domain in which identity, payoff, residual, and evidence invariants can be falsified without hiding irreducible path dependence.

5.2 Contract normalization

A source contract is lowered through an explicit, versioned normalization profile. The normalized record binds:

- source venue namespace and contract identifier;
- source terms digest, revision, and observation time;
- economic reference namespace, identifier, unit, administrator, methodology, methodology digest, and fallback;
- calendar, timezone, observation window, sampling, disruption, and correction;
- comparator and terminal payoff function;
- settlement asset, scale, rounding, deadline, dispute, default, governing-rule reference, and finality domain;
- explicit non-equivalences and unsupported terms.

The normalized value must be opaque to callers until validation succeeds. Missing material terms reject lowering. The system must never “fill in” a benchmark, timezone, settlement source, or rounding rule because a similarly named product usually uses one.

Illustrative pseudo-code:

```
validated = validate_and_lower(source_terms, normalization_profile)
state_receipt = derive_state_key(validated)
residual = analyze_residual(target, candidate_portfolio, state_domain)
completeness = assess_completeness(residual, books, clearing, settlement, evidence)
```

This is an interface contract, not implemented code.

5.3 State identity

The StateKey is a versioned digest over the canonical semantic preimage, not an asset ticker or free-form label:

```
StateKey = H(
 domain_tag,
 schema_version,
 reference_identity,
 methodology_and_fallback,
 calendar_timezone_and_observation,
 comparator_and_payoff,
 disruption_and_correction,
 settlement_asset_scale_rounding,
 deadline_dispute_default_and_finality
)
```

Set-like fields require canonical ordering. Material term changes change the key. Equivalent source serialization does not. Golden preimages and an independent checker are required before cross-language use. The repository's existing digest helpers are useful precedents but are not automatically a cross-language StateKey format.

5.4 Payoff and residual algebra

Let `Omega` be the declared set of admissible terminal states, `A` the set of settlement assets, and `p_i(omega)` the exact asset-vector payoff of contract `i` in state `omega`. For target payoff `g` and rational quantities `q_i`, the financial residual is:

```
r(omega) = g(omega) - sum_i q_i p_i(omega)
```

Exact payoff replication requires `r(omega) = 0` for every supported state and no unsupported state. Approximate replication reports, rather than hides:

- the evaluated domain and its blind spots;
- worst-case and scenario residuals;
- basis, jump, timing, FX, default, legal, and liquidity residual classes;
- coefficient, quantity, price, fill, and valuation assumptions;
- source time and digest for every observation;
- sensitivity to model and benchmark changes.

Asset conversion uses conservative upper-bound valuation and declared scales. Overflow, missing prices, stale prices, zero denominators, unrecognized assets, or ambiguous rounding fail closed. Rounding never increases an allowed release.

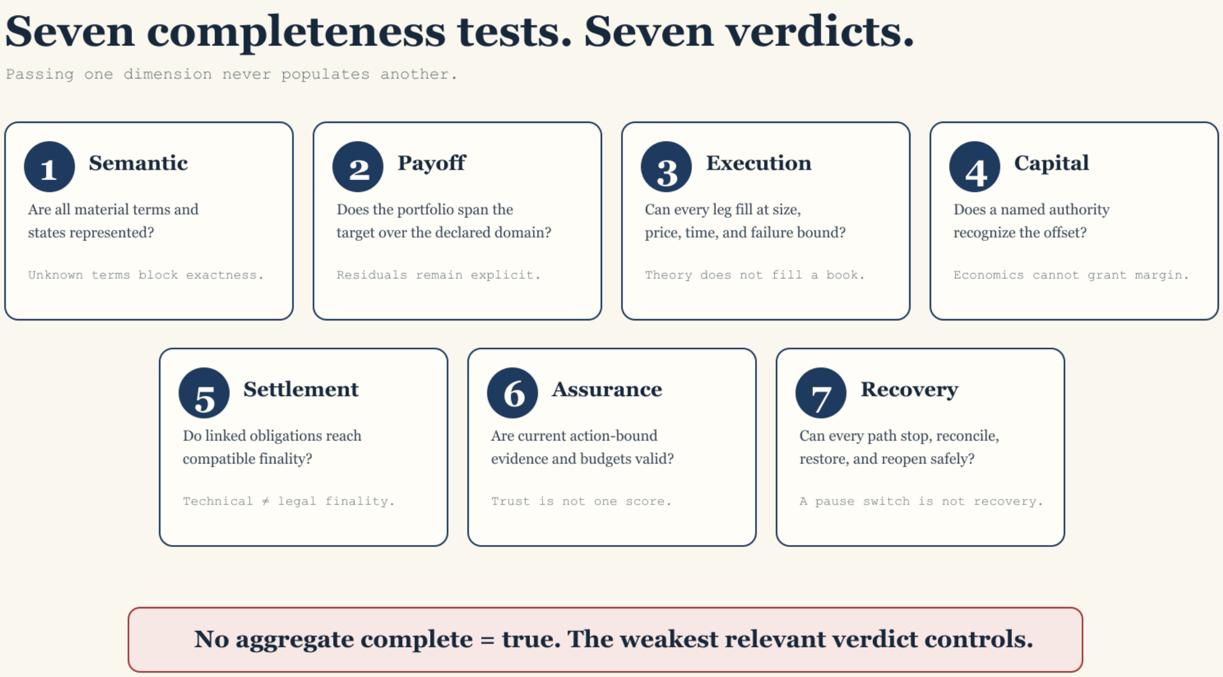
5.5 Perpetual profiles remain path dependent

A perpetual is not a terminal claim without an expiry. Its cash flows depend on the entire path of marks, funding, collateral, liquidation, oracle availability, pauses, position management, and exit timing [T07-T09]. Statebook should model a perpetual as a continuing hedge profile beside terminal claims. Any conversion to a terminal exposure requires an explicit close, roll, liquidation, and funding model. The remaining path residual stays visible.

5.6 Seven typed completeness dimensions

The initial three-part distinction — product, execution, and capital completeness — is necessary but not sufficient. The full decision requires seven typed dimensions:

1. Semantic completeness: Are all material terms and relevant states represented without unknowns?
2. Payoff completeness: Does the candidate portfolio span the target payoff over the declared domain within the stated tolerance?
3. Execution completeness: Can every required leg fill at the claimed size, price, time, and failure bound?
4. Capital completeness: Does the governing margin or clearing authority recognize the offset before liquidation, with explicit haircut and model?
5. Settlement completeness: Can linked obligations reach compatible technical, economic, and legal finality?
6. Assurance completeness: Are current evidence, authorization, provenance, replay, solvency, route, and anomaly requirements satisfied for the proposed action?
7. Recovery completeness: Can every externalization path stop, every in- flight item reconcile, evidence remain available, liabilities restore without duplication, and the system reopen through bounded canary stages?



Seven completeness tests · explanatory media, not evidence

There is no aggregate `complete = true`. Each dimension returns a typed status, evidence references, missing facts, and residuals. The weakest relevant dimension controls the permitted claim and action.

6. HSAI and repository integration boundary

This repository already contains useful evidence and assurance patterns:

- `hsai-claim-envelope` distinguishes evidence maturity, predicates, trust roots, provenance, conjunction, and acceptance policy.
- `hsai-agent-case` separates declared input from evidence lanes.
- `hsai-attestation` and `hsai-attestation-phala` demonstrate injected verifiers, caller-supplied trust roots, normalized outputs, replay protection, hermetic doubles, and `Attested`-only claims.
- `hsai-agent-admission` at committed revision `b4b644cd96d9b70eb21ff6681a0014245773cd0f` demonstrates proposal, decision, reason, digest, quarantine, journal, replay, and no-authority patterns. The current dirty working tree remains outside this slice and is not a stable Statebook seam.
- `zkbench-core` demonstrates Semantic IR, typed completeness, evidence candidates, acceptance, escalation guards, quarantine, deterministic artifacts, and explicit Claim Boundaries.

The financial and assurance algebras must stay separate:

```
Financial algebra:
 terms -> normalized contract -> StateKey -> payoff -> residual -> completeness

Assurance algebra:
 observations -> appraisal -> current evidence -> policy decision -> claim limit
```

HSAI can attest that decision digest `D` passed policy `P` against evidence snapshot `E` at time `T`. It cannot thereby prove market price, payoff equivalence, venue solvency, legal enforceability, settlement finality, or semantic correctness. A higher evidence maturity for the wrong property cannot substitute for the missing property.

The eventual integration should use a narrow adapter. Financial semantics are computed before the adapter. The adapter preserves missing information as `Unknown` and never receives settlement authority. Admission receives a completed, digest-bound Statebook proposal; it must not reconstruct economic meaning from a lossy target string and integer amount.

7. Assurance-adjusted externalization

7.1 The security problem is path coverage

“Kill instant settlement” captures a real intuition: irreversible value release can turn one incorrect state transition into final loss before detection and intervention. It is not a sufficient architecture. A mandatory delay protects only the paths it actually gates. If profitable closes, collateral withdrawal, bridges, administrative payouts, or internal credit reuse bypass the delayed withdrawal path, one delayed function creates the appearance rather than the substance of containment.

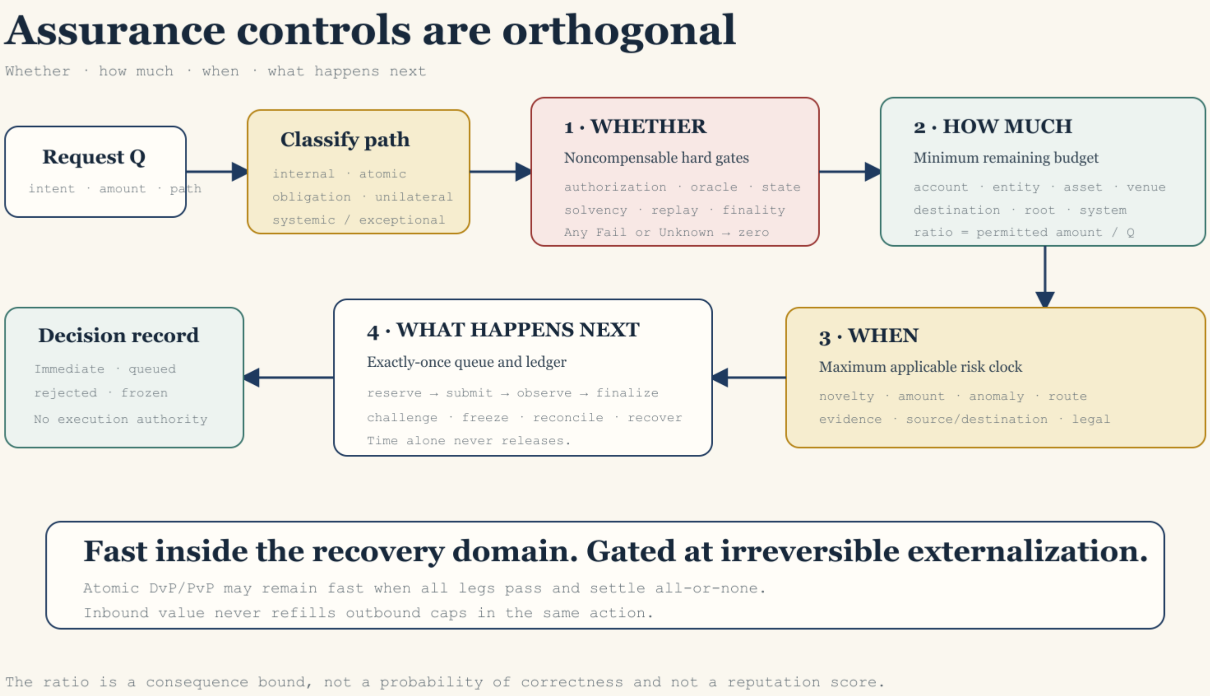
The correct scope is every transition that can convert disputed, newly created, or weakly supported state into spendable value, borrowing capacity, reusable margin, transferable claims, or external assets.

7.2 Three clocks

The architecture separates:

1. Execution finality: when a match or state transition becomes irrevocable inside the market engine.
2. Economic settlement: when balances, collateral, and obligations update inside the recovery domain.
3. Externalization finality: when value leaves the domain in a way the system cannot reliably stop or reverse.

Valid internal risk updates should generally remain fast. Delaying mark, funding, collateral lock, liquidation, or loss recognition can make a system less safe. Atomic DvP or PvP should remain available where it truly guarantees all legs or none. The highest friction belongs on unilateral irreversible externalization whose evidence or consequence is uncertain.



Assurance-adjusted settlement controller · explanatory media, not evidence

7.3 Why there is no trust score

A single ratio between “trust” and “instant settlement” is unsafe. It lets strong identity or history compensate numerically for a failed oracle, stale authorization, missing solvency evidence, or compromised signer. It also double-counts correlated sources and turns hidden weights into financial authority.

The retained controller is lexicographic, not compensatory:

1. Hard gates determine whether. Every applicable critical predicate must pass. `Unknown` is not `Pass`.
2. Exposure budgets determine how much. Native-asset and conservative common-numeraire caps bind account, subject, destination, StateKey, oracle root, route, dependency, venue, class, and system exposure simultaneously.
3. Risk clocks determine when. The required delay is the maximum applicable delay, not a weighted average.
4. The queue determines what happens next. Time alone never converts an invalid request into a valid release.

Reputation can select less friction only inside the current hard envelope. It cannot bypass it.

7.4 Release classes

The controller classifies a transition before calculating a release:

- `InternalRiskState`: valid bookkeeping, marks, funding, collateral locks, or liquidation state that remains inside the recovery domain.
- `AtomicLinkedExchange`: an all-or-none, exactly-once exchange in which every leg is bound, prefunded, unencumbered, within gross caps, and subject to compatible finality. Inbound consideration never replenishes outbound caps in the same operation.

- `ExternalRiskReducingObligation`: a fixed-beneficiary, fixed-amount, restricted, prefunded obligation whose independently recomputed before/after exposure is strictly lower and whose destination cannot be used as a general withdrawable balance.
- `ExternalUnconditional`: unilateral value release subject to ratio, caps, and queue.
- `SystemicOrExceptional`: novel, concentrated, disputed, or high-impact state receiving zero ordinary instant release.

Unknown classification becomes `SystemicOrExceptional`. A class changes timing and obligation structure. It never creates an uncapped bypass.

7.5 Hard gates

Before any instant amount is considered, the system requires current, request-bound evidence for:

- action and destination authorization;
- source authenticity, freshness, monotonicity, one-time use, and equivocation control;
- independent recomputation of PnL, collateral, redemption, or transfer;
- existence and exactly-once identity of the originating state transition;
- reserves, liquid resources, and declared stress support;
- destination, sanctions, route, bridge, and contract-behaviour policy;
- absence of a relevant anomaly, freeze, rollback, compromise, or exhausted system loss budget;
- independence of required evidence roots;
- terms digest, StateKey, coherence, and settlement-report binding for contract-derived value;
- zero reuse of pending or anomalous PnL;
- exact obligation or linked-plan structure where an expedited class is used.

Any required `Fail` or `Unknown` yields zero instant externalization. The ordinary queue is not a delayed permission to release invalid value.

Pending or anomalous PnL remains non-reusable until an explicit `ReuseFinality` predicate passes:

```
ReuseFinality = AND(
 source_finality_profile_passes,
 payoff_and_PnL_are_independently_recomputed_and_economically_final,
 operational_reconciliation_and_availability_pass,
 applicable_legal_entitlement_and_reversal_rules_pass,
 no_live_challenge_anomaly_freeze_or_solvency_failure
)
```

Technical inclusion or source finality alone never unlocks withdrawal, borrowing, bridging, transfer, margin reuse, or release-budget value. An unknown or inapplicable-but-undeclared dimension fails closed.

7.6 Dynamic release surface

For request amount `Q`, conservative value `V(Q)`, current assurance tier `t`, and applicable remaining caps `B_j`, the candidate instant amount is bounded by:

`Q` must be strictly positive. Direction is encoded separately from amount. Zero, negative, wrong-sign, overflowed, or noncanonical amounts reject before valuation, gate evaluation, reservation, or ratio calculation.

```
if any required gate != Pass:
 instant = 0
else:
 common_limit = min(V(Q), tier_cap[t], B_1, B_2, ..., stressed_system_limit)
 native_limit = floor_convert(common_limit)
 instant = min(Q, native_asset_cap, native_limit)

instant_ratio = instant / Q
```

This ratio is an output, not a reputation input. It changes with current evidence, amount, destination novelty, correlated exposure, system stress, route, instrument, and policy version. The same actor can receive different results for different actions at the same time.

If `0 < instant < Q`, the request is partitioned into independent parts:

```
ReleaseRequest {
 request_id,
 total_amount,
 release_parts
}

ReleasePart {
 part_id,
 amount,
 queue_status,
 transfer_status,
 reservation_id?
}
```

Every instant or queued part has an independent id and lifecycle. No amount may belong to two parts. Consumed, reserved, in-flight, queued, cancelled, and remaining amounts reconcile exactly to the parent amount. `reservation_id` is absent while transfer status is `Unreserved` and required for every later transfer status, including `ProvenNoOutflow`, so the released reservation remains auditable.

Atomic linked plans and risk-reducing obligations are all-or-none at the validated amount. They never degrade into unilateral legs. Splitting one request across accounts, destinations, blocks, assets, agents, or correlated venues cannot increase aggregate allowance.

7.7 Exactly-once exposure ledger

All applicable counters are checked and reserved in one compare-and-swap transition against a versioned ledger tip. A request and reservation are exactly-once. State is:

```
remaining = limit - consumed - live_reservations - in_flight
```

An immediately final release moves reservation to consumed. A submitted cross-domain release remains `in_flight` until destination finality is observed. Destination finality atomically moves exposure from `in_flight` to `consumed`; it does not restore capacity. Consumed exposure continues to reduce capacity until the deterministic refill transition. Timeout or reported failure removes a reservation or in-flight debit only with independently validated proof that no value left. Otherwise exposure remains counted. There is no generic cancel- and-release command.

Exactly-once identity is bound by a canonical, domain-separated digest:

```
intent_digest = H(canonical_encode(
 domain_tag,
 schema_version,
 subject,
 source_account,
```

```
 destination,
 route,
 asset,
 direction,
 total_amount,
 StateKey_or_financial_basis,
 originating_transition,
 linked_plan_or_obligation_digest,
 action_authorization_digest,
 declared_release_class,
 nonce,
 requested_at,
 expires_at
))

decision_context_digest = H(canonical_encode(
 intent_digest,
 evidence_snapshot_digest,
 valuation_profile_digest,
 policy_digest,
 evaluated_at
))

release_attempt_digest = H(canonical_encode(
 release_part_id,
 decision_context_digest,
 reservation_id
))
```

Golden vectors independently verify every encoding and digest. The same intent id with a different immutable `intent_digest` rejects before reservation. Fresh evidence, valuation, or policy creates a new decision context and release attempt linked to the same parent intent. Only one live or submitted attempt may exist per release part. Retries preserve the same external idempotency key.

Refill is a deterministic, capped, sequential epoch transition. It requires finality, reconciliation, independently closed correlated exposure, clean epochs, and absence of circular or pending-credit origin. A missed evidence epoch cannot be backfilled later using today's clean signal.

7.8 Challenge queue and hysteresis

```
QueueStatus =
 None | Queued | Challenged | EvidenceExpired |
 Frozen | Cancelled | RevalidationRequired

TransferStatus =
 Unreserved | Reserved | Submitted |
 SourceObserved | SourceFinalized |
 DestinationObserved | DestinationFinalized |
 Consumed | ProvenNoOutflow
```

These statuses apply per `ReleasePart`. Every release after waiting requires a fresh decision context and fresh budget reservation. A revalidated queued part transitions to `Reserved`, never directly to “released.” Legal finality remains a separate attached status. Same-domain state coalescing requires an explicit finality adapter. Pending amounts provide zero borrowing, collateral, margin- reuse, transfer, or release-budget value. Challenges use a bounded evidence grammar, not arbitrary vetoes.

Valid combinations are constrained. `Queued`, `EvidenceExpired`, `Cancelled`, and `RevalidationRequired` require `Unreserved` and no reservation id. `Challenged` and `Frozen` block new reservation and submission side effects. They never block observation, source or destination finality, no-outflow proof, or reconciliation transitions for an already submitted transfer. `Consumed` and `ProvenNoOutflow` require `QueueStatus::None`, retain the reservation id, and are terminal for that part.

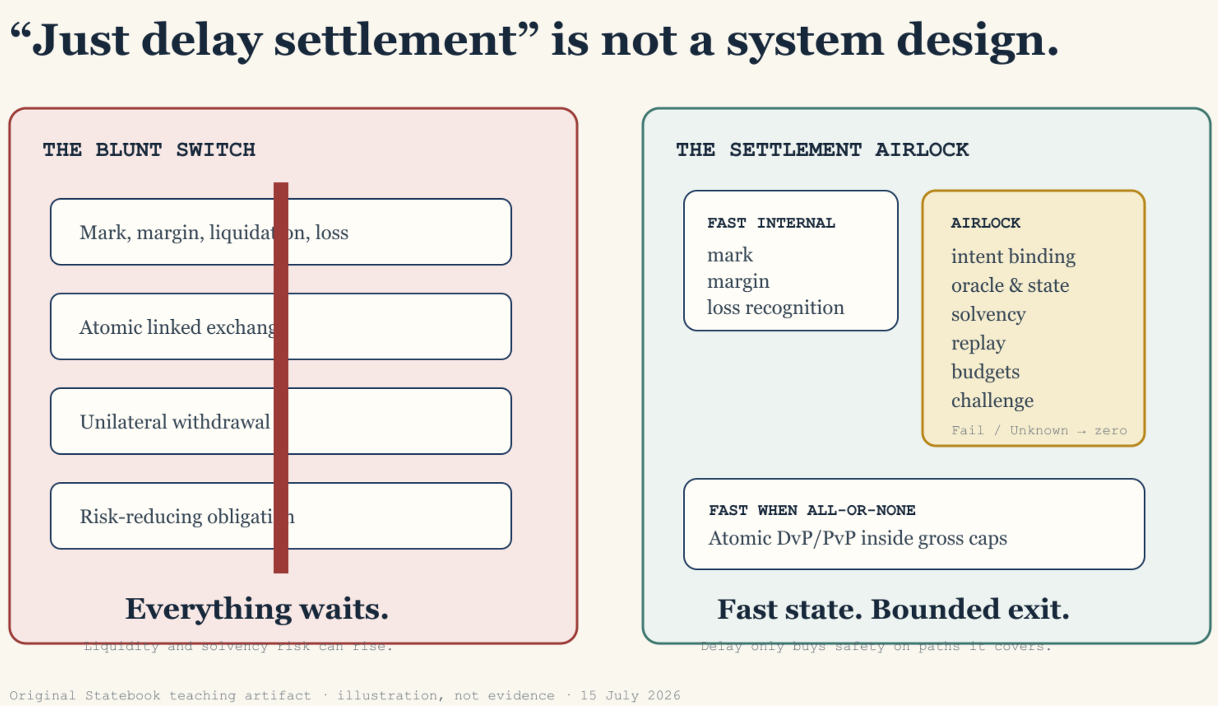
Risk tightens faster than it relaxes. New adverse evidence can reduce caps or lengthen delays immediately. Relaxation requires resolution evidence, independent current support, clean epochs, dwell time, timelock, retained- traffic shadow evaluation, and a new policy digest. Emergency authority may tighten a bounded lane. It cannot silently increase instant release.

Breaker progression is explicit:

```
Normal -> Guarded | Halted
Guarded -> Challenged | Halted | Resolution
Challenged -> Guarded | Halted | Resolution
Halted -> Resolution
Resolution -> Recovery
Recovery -> Normal | Guarded | Halted
```

TTL or cumulative-renewal exhaustion enters `Resolution`. It neither releases disputed value nor permits silent renewal or indefinite limbo. Entitlement- based safe exits, insolvency priority, adjudication authority, and pending-obligation treatment are declared by policy. There is no direct `Halted -> Normal` transition.

7.9 Control tradeoff



Settlement-delay meme · explanatory media, not evidence

A delay buys a detection and intervention option. It also creates liquidity, censorship, custody, governance, and false-positive costs. The objective is not maximum delay. It is minimum irreversible exposure consistent with market function, bounded by the slowest credible detection-and-response clock for the specific action.

8. Incident-grounded lesson: Ostium, July 2026

8.1 What was observed

Contemporaneous reporting quoted an Ostium X post as stating that the project was aware of an OLP-vault issue and had paused all trading. The original post was not independently retrievable or retained during this review; this is reporting of a first-party statement, not a retained first-party artifact [S07]. The cited Arbitrum transaction and a preliminary Blockaid analysis supplied additional early context [S09-S10]. Public documentation described its oracle, automation, immediate trader-PnL, and LP-withdrawal systems [S11-S12]. Pinned public source paths expose one revision's profitable-close, vault- transfer, authorization, upkeep, and timestamp checks [S13-S14].

Those materials establish a bounded design lesson, not a final root-cause finding. The source commit is not asserted to match incident-time deployed bytecode. One reported official follow-up URL was not independently retrievable during the second-pass review and is not used for a distinct substantive claim [S08]. Preliminary social posts may be superseded. The exact loss, recovery, and counterfactual effect of any delay remain outside this paper's evidence.

8.2 What the incident prompt teaches

The salient question is not “did the protocol have a withdrawal delay?” It is “which paths turned state into externalizable value, and did each path require current, independently recomputed, request-bound evidence before finality?”

Current Ostium documentation distinguishes a dynamic liquidity-provider request-and-settle withdrawal flow, typically described as taking days, from winning-trader PnL paid immediately onchain. That distinction supports the path- coverage lesson: a delayed LP redemption route does not constrain a separate profitable-close payout route.

Direct event-log inspection of the cited Arbitrum transaction shows ten price requests and responses and five open-and-close round trips in one successful transaction. That establishes an anomalous execution pattern. It does not prove who compromised what. Preliminary Blockaid analysis alleged involvement of a registered forwarder and prepared or future-dated authorized reports. The pinned router code rejects request timestamps that are still later than the current block, so this paper does not restate the preliminary allegation as acceptance of a still-future request timestamp.

The correct counterfactual is conditional:

- A challenge window could reduce loss only if the compromised or erroneous path entered the window.
- Detection and intervention had to occur before the window closed.
- Pending credit could not be borrowed, transferred, bridged, or reused.
- Emergency controls had to remain available and uncompromised.
- The system still needed safe margin, liquidation, and loss-recognition paths.

Therefore this paper does not claim that mandatory delays would have prevented the incident. It claims that irreversible path coverage is a first-class security invariant.

8.3 Wider incident evidence

Nomad illustrates how one validation defect can become a copyable machine-speed loss path [S15]. Bybit's 2025 disclosures and preliminary Verichains analysis show how interface and signing-chain compromise can defeat the intuition that a cold wallet is safe merely because its keys are offline [S16-S17]. These incidents are architecturally different. Together they demonstrate that “trusted component” is not a stable security primitive. Every high-consequence action needs current, intent-bound, independently observable controls.

9. Threat model and control coverage

9.1 Assets

The system must protect:

- user collateral and settlement assets;
- contract terms and StateKey integrity;
- market, oracle, and benchmark observations;
- margin, exposure, and budget state;
- queued rights and challenge evidence;
- policy, governance, and trust-root configuration;
- signer, relayer, venue, bridge, and finality identities;
- audit records and nonclaims.

9.2 Adversaries and failures

The design assumes possible:

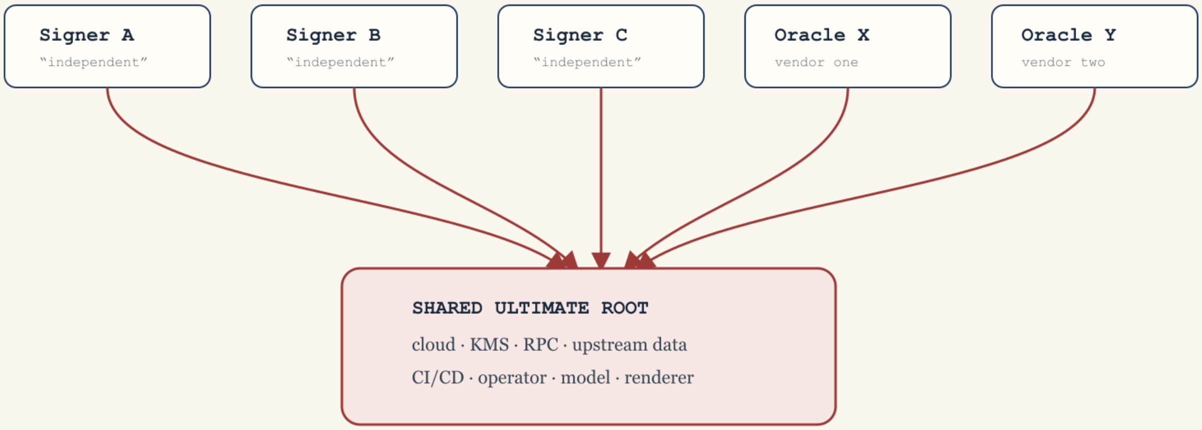
- compromised trader, operator, signer, relayer, oracle, benchmark, bridge, venue, evidence provider, or governance key;
- malicious or mistaken contract normalization;
- stale, replayed, equivocated, correlated, or transplanted evidence;
- model, arithmetic, scale, rounding, and valuation defects;
- partial linked-leg execution;
- venue insolvency, withdrawal freeze, chain reorganization, or legal stay;
- queue censorship, challenge spam, cap splitting, and policy oscillation;
- AI-assisted exploit search, malformed contract generation, social engineering, or coordinated machine-speed withdrawal.

9.3 Control map

Failure domain	Primary Statebook control	Residual risk
Semantic mismatch	Terms digest, normalization profile, StateKey, unsupported-state reporting	Wrong or incomplete source terms
Oracle or benchmark compromise	Multiple current roots, request binding, freshness, equivocation checks, caps, queue	Correlated or governance-level compromise
Signer compromise	Intent-bound payload, destination policy, caps, novelty delay, independent watchers	Compromise of all required roots or governance
Calculation defect	Independent deterministic recomputation, exact arithmetic, golden vectors	Shared specification bug
Replay or concurrency	Exactly-once ids, CAS ledger tip, append-only records	Availability loss under contention
Linked-leg failure	All-or-none plan, gross outbound budgets, no inbound netting	External domain violates claimed atomicity
Bridge or finality reversal	In-flight accounting until observed finality, no timeout release	Prolonged trapped exposure
Insolvency or liquidity stress	Reserve/liquidity gates, stressed system budget, no pending-credit reuse	Evidence can lag sudden insolvency
Governance attack	Fast bounded tightening, slow timelocked relaxation, scoped breakers	Malicious threshold coalition or liveness failure
AI swarm	Aggregate multidimensional caps, split resistance, queue priority controls	Novel correlated tactics and denial of service

Five vendors. One failure root.

Independence is a dependency graph, not a headcount.



Quorum count: 1 until the shared root is separated.

Three signatures on one compromised screen are one control domain.

Original Statebook teaching artifact · illustration, not evidence · 15 July 2026

Correlated dependency roots · explanatory media, not evidence

9.4 What delays do not solve

Delay does not repair a false oracle, replace solvency, validate legal rights, prevent a compromised governance majority, create liquidity, guarantee watcher availability, or make a bridge final. It can worsen margin pressure and concentrate discretionary control. The queue is one control in a layered system, not a universal security theorem.

10. Capital and legal coherence

10.1 Economic offset is not recognized offset

Suppose two positions have nearly opposite payoffs. The holder may still post gross collateral if they sit in separate accounts, entities, customer protection regimes, default funds, currencies, legal jurisdictions, or close-out netting sets. Duffie-Zhu's result means even centralization can reduce one netting benefit while destroying another [I06].

Capital completeness must name:

- the authority recognizing the offset;
- eligible accounts and legal entities;
- model version, stress scenarios, correlations, concentration add-ons, and haircuts;
- collateral eligibility and FX treatment;
- liquidation and default-waterfall rules;
- enforceable close-out netting and insolvency treatment;
- conditions under which the recognition is withdrawn.

A Statebook may calculate an economic residual. It cannot award margin relief unless the governing authority adopts it.

10.2 Technical and legal finality

A ledger can mark a transfer final while law still permits reversal, stay, or insolvency challenge. A payment can be irrevocable in one domain while a linked asset leg remains conditional in another. Statebook settlement reports must distinguish technical confirmation, economic availability, operational reconciliation, and legal finality.

10.3 Regulatory topology

Product convergence crosses regulatory categories rather than dissolving them. Event contracts, futures, swaps, options, securities, gaming, insurance, payments, commodities, and lending can be treated differently by jurisdiction and participant type. Benchmark governance, market integrity, surveillance, customer protection, segregation, best execution, disclosures, sanctions, privacy, and data rights remain separate obligations.

The Statebook should carry legal and authority references as constraints and unknowns. It should never infer legal fungibility from payoff similarity.

11. AI, technology, and the global economy

11.1 AI creates both underlyings and market actors

AI affects Statebook through four channels:

1. Physical demand: data centres consume electricity, grid capacity, cooling, land, networking, chips, transformers, turbines, and capital [A01, A04].
2. Economic productivity: automation changes costs, labour demand, competition, investment, and sectoral output unevenly [A05, A05A, A05B, A06, A07].
3. Financial infrastructure: models improve normalization, surveillance, scenario generation, code analysis, and routing, while creating correlated model and third-party risk [A08, A09, I15].
4. Adversarial capacity: agents accelerate reconnaissance, exploit replication, transaction splitting, synthetic identity, persuasion, and congestion. This is a threat-model assumption, not an empirical forecast.

The same system that helps discover a hedge can discover a bypass. Proposal agents cannot mutate state; they may only suggest mappings, portfolios, tests, or transactions. A separately identified and validated anomaly detector may emit evidence only. Deterministic pre-authorized policy may map that evidence to a fixed-scope, fixed-TTL protective transition. No model selects caps, scope, TTL, renewal, policy, release, challenge resolution, or reopening.

11.2 Physical and institutional bottlenecks

The IEA's 2026 central scenario places data-centre electricity use near 950 TWh in 2030, roughly double the 2025 level, while emphasizing grids, equipment, chips, capital, and social acceptance as constraints [A01]. World Bank work shows that connectivity, compute, context, and competency are distributed unevenly [A04]. FERC's 2026 large-load action illustrates how AI demand becomes a grid-governance problem [A04A]. These constraints create potential hedging demand but also make references locational, methodological, and political.

11.3 Productivity is a distribution, not one number

OECD scenarios estimate material annual labour-productivity gains in highly exposed G7 economies under explicit assumptions [A05]. Field evidence such as Brynjolfsson, Li, and Raymond finds large gains in a specific customer-support setting [A05A]. Acemoglu's task framework produces a more restrained aggregate baseline [A05B]. The correct interpretation is not to choose the most exciting number. It is to model adoption cost, task exposure, complementarity, diffusion, competition, energy, and institutional response as uncertain state variables.

11.4 Macro crosscurrents

The IMF's July 2026 baseline projects 3.0 percent global growth in 2026 and 3.4 percent in 2027, balancing energy and geopolitical downside against technology investment [A02]. The World Bank's June 2026 baseline is more conservative at 2.5 percent for 2026 and includes a severe downside scenario near 1.3 percent [A03]. These are not numbers to average mechanically. They use different cutoffs, assumptions, and scenario structures.

Statebook-relevant feedback loops include:

- AI capital expenditure raises chip, power, construction, and financing demand.
- Grid and supply constraints raise regional basis risk.

- Higher productivity can improve fiscal capacity but can also concentrate rents and widen country or firm divergence.
- Sovereign debt, energy shocks, and geopolitical fragmentation can tighten funding precisely when infrastructure investment needs rise.
- Tokenization and 24/7 markets can accelerate collateral movement while increasing operational and liquidity demands.
- Common AI, cloud, oracle, and data providers create correlated failure roots.
- Automated risk reduction can become synchronized deleveraging.

11.5 Progress occurs at different clocks

Software capability can improve weekly or monthly. Hardware capacity expands over quarters and years. Transmission grids, power plants, permitting, standards, clearing arrangements, and law move over years or decades. A coherent market substrate must expose these mismatched clocks rather than assuming that software speed removes physical or legal latency.

11.6 Two-, five-, and ten-year scenario envelope

These are scenarios, not forecasts.

2026-2028: normalization before unification

Likely developments:

- more venues add unfamiliar perpetual, event, tokenized, compute, and energy references;
- institutional and onchain products overlap in user intent but retain separate collateral and legal domains;
- AI assists contract ingestion and anomaly detection, with deterministic review still required;
- demand grows for shared contract semantics and provenance;
- security policy starts distinguishing fast internal state from delayed externalization;
- regulatory classification and benchmark governance remain product-specific.

The practical Statebook opportunity is read-only: normalize terms, find candidate relations, expose residuals, and generate auditable completeness reports.

2029-2031: federated execution and selective capital recognition

Plausible developments:

- compute and energy benchmarks become more standardized but remain locational;
- cross-venue intent routing and linked settlement mature for bounded asset pairs;
- some clearing or prime-broker arrangements recognize selected cross-product offsets;
- policy engines use current evidence to govern externalization caps and challenge windows;
- AI agents become major proposal and liquidity actors but remain bounded by explicit authority and loss budgets;
- correlated infrastructure and model providers become a major prudential concern.

The product challenge shifts from mapping to governed federation. Statebook can propose linked portfolios, but recognized capital and finality remain authority- specific.

`2032-2036: market substrate or systemic monoculture`

Two paths are plausible:

- Coherent federation: open semantic standards, portable evidence, diverse execution, interoperable settlement, explicit legal mappings, and bounded assurance policies let new risk markets compose without one universal venue.
- Systemic monoculture: a few cloud, model, oracle, clearing, and settlement providers dominate. Apparent efficiency masks correlated governance, operational, cyber, and liquidity failure.

The difference is not whether infrastructure becomes programmable. It is whether semantic and authority boundaries remain inspectable as programmability scales.

12. Product architecture

12.1 Layers

The proposed product has six layers:

1. Source layer: venue terms, standards, benchmarks, legal references, market observations, and evidence, each with provenance and version.
2. Semantic layer: validated terminal contracts, StateKeys, payoff functions, and explicit non-equivalences.
3. Risk layer: residual analysis, typed completeness, conservative valuation, and scenario stress.
4. Assurance layer: current evidence resolution, hard gates, linked-plan and obligation validation, exposure budgets, queue, and audit decisions.
5. Recovery layer: all-path stop, in-flight reconciliation, immutable evidence, restored liabilities, root rotation, incident replay, and canary reopening.
6. Authority layer: separately governed execution, clearing, custody, pause, signing, and settlement systems. This layer is outside the first product.

12.2 Proposed modules

- `statebook-core`: financial terms, StateKey, exact payoff and residual, semantic and payoff completeness, and the common typed-result vocabulary.
- `statebook-settlement`: execution, capital, settlement, assurance, and recovery completeness; assurance observations; valuation; linked plans; obligations; budgets; queue; recovery; and one transition kernel. The kernel composes all seven results without allowing either module to fabricate the other's evidence.
- `statebook-hsai`: narrow evidence and decision-envelope adapters.
- `statebook-sim`: deterministic clocks, scenarios, attacker actions, sweeps, and counterexamples.
- `statebook-report`: digest-bound manifests, readback validation, nonclaims, and portable audit bundles.

These are future module boundaries, not currently authorized crates. The first code phase requires a separately approved state slice.

12.3 Deep interface: one settlement transition kernel

Queue and budget mutations must not be coordinated by callers. One kernel owns the atomic state transition:

```
decide_and_transition(request, current_state)
 -> rejected | quarantined | immediate | queued | frozen
```

Its output binds contract, residual, completeness, evidence, policy, valuation, linked-plan or obligation, ledger before/after tips, queue transition, reasons, and nonclaims. The output is a decision record. It is not an execution command.

12.4 Evidence and media registry

Every external item records stable URL or commit, author or publisher, publication and retrieval time, digest when captured, license or quotation basis, evidence class, supported claims, limitations, and admissibility. Tweets, screenshots, podcasts, and memes default to `IllustrativeNarrative`. They never influence price, assurance, solvency, finality, or settlement unless promoted through a separately specified capture and review process.

The SVG diagrams and memes in this publication are original explanatory assets. They contain no copied commercial meme template or third-party logo.

13. Business and governance model

13.1 Likely users

- traders and treasury teams seeking cross-product hedges;
- venues and brokers seeking semantic routing and product discovery;
- clearing and margin teams evaluating bounded offset recognition;
- risk, legal, and compliance teams reviewing basis and finality;
- protocol and exchange security teams governing externalization;
- researchers and regulators inspecting new frontier-risk markets;
- AI agents proposing, but never authorizing, structured exposures.

13.2 Product wedge

The lowest-risk commercial wedge is a read-only coherence terminal and API:

- ingest a bounded set of source terms;
- show normalized differences;
- compare terminal payoffs;
- produce exact or approximate residual reports;
- display all seven completeness dimensions;
- simulate assurance-adjusted externalization policies;
- export an audit bundle.

This creates value before custody, execution, margin, or settlement authority. Later products can add venue adapters, intent routing, or policy integration only after separate evidence and authorization.

13.3 Economic model

Potential revenue surfaces include enterprise data and API subscriptions, normalization and benchmark-governance tooling, simulation and audit products, venue integration, and separately governed execution-intent fees. The business must avoid incentives that reward labeling more portfolios as equivalent or more releases as safe. Accuracy, explicit unknowns, and counterexample discovery are product outputs, not friction to hide.

13.4 Governance

Governance separates:

- schema and semantic-profile changes;
- benchmark and source-root changes;
- valuation and risk-model changes;
- settlement policy and cap changes;
- emergency tightening;
- authority integrations.

Every change has a digest, effective time, review record, rollback rule, and claim boundary. Emergency paths may narrow or freeze a bounded lane. Expansion of value release requires timelock and evidence.

14. Implementation sequence

Phase 0 — publication boundary

This paper, the source index, PRD, media, and verified PDFs. No runtime capability.

Phase 1 — terminal semantic fixtures

Define bounded scalar cash-settled fixtures, exact quantities, source digests, normalization profiles, StateKey golden vectors, and negative cases. No live adapters.

Phase 2 — payoff and residual engine

Implement exact rational terminal payoff comparison over finite declared state domains. Report unsupported states and residual classes. No execution.

Phase 3 — completeness reports

Add hermetic book, clearing, settlement, evidence, and recovery fixtures. Return seven typed dimensions. No global boolean and no capital recognition claim.

Phase 4 — deterministic settlement simulation

Implement assurance resolution, conservative valuation, multi-axis exactly-once budgets, linked-plan validation, obligation validation, queue transitions, hysteresis, and adversarial scenarios under an injected clock. No value moves.

Phase 5 — evidence adapters and audit bundles

Add narrow HSAI and fixture adapters, digest-bound decision records, portable reports, readback validation, and explicit nonclaims. No live venue authority.

Phase 6 — read-only external adapters

After separate authorization, ingest live or captured venue terms and market data without trading. Validate source identity, version, schema, and provenance.

Phase 7 — separately owned authority integration

Execution, custody, clearing, pause, signing, margin, or settlement requires a new threat model, legal review, operational evidence, loss limits, and authority boundary. It is not implied by completing prior phases.

15. Evaluation and falsification

A Statebook thesis is useful only if it can fail visibly. Required evaluation includes:

- semantic-equivalence precision and recall on a reviewed fixture corpus;
- independent StateKey golden-vector agreement;
- exact residual recomputation and permutation invariance;
- unsupported-state detection rate;
- execution-feasibility error under retained book snapshots;
- false capital-recognition rate, whose target is zero;
- hard-gate bypass count, whose target is zero;
- pending-PnL reuse count, whose target is zero;
- cap split-resistance across account, asset, destination, route, and venue;
- exactly-once ledger behavior under concurrency;
- partial linked-leg execution count, whose target is zero in the model;
- queue release without fresh evidence count, whose target is zero;
- false-positive freeze duration and legitimate-liquidity cost;
- incident detection, validation, and intervention latency distributions;
- counterexample minimization and deterministic replay;
- report readback integrity and claim-boundary compliance.

Rejection criteria include:

- term normalization cannot reach acceptable reviewed accuracy;
- equivalence false positives cannot be bounded;
- the residual model depends on unobservable or unstable state;
- exposure budgets can be expanded by splitting or correlated roots;
- controls materially increase insolvency or liquidation risk;
- operators cannot explain or reproduce decisions;
- legal or clearing authorities cannot use the semantic representation;
- the business requires hiding unknowns or overstating safety.

No production threshold or parameter is selected in this paper.

16. Required invariants

1. Every mutation names the exact state slice it touches.
2. Source terms remain digest-bound to every normalized contract.
3. Material term differences cannot share a StateKey.
4. Unsupported states cannot produce `Exact` payoff status.
5. Perpetual path dependence cannot disappear through terminal relabeling.
6. No completeness dimension implies another; specifically, payoff completeness does not imply semantic, execution, capital, settlement, assurance, or recovery completeness.
7. Missing observations remain `Unknown` or `NotObserved`, never `Pass`.
8. A higher evidence maturity cannot substitute for the wrong property.
9. Any failed or unknown applicable hard gate yields zero instant release.
10. Pending anomalous value supplies zero collateral, borrowing, margin-reuse, or release capacity.
11. All externalizing classes consume applicable gross native and common caps.
12. Inbound linked legs do not replenish outbound capacity in the same action.
13. A linked plan executes all legs or none.
14. A risk-reducing obligation cannot become a general withdrawable balance.
15. Ledger mutation is exactly-once against one expected tip.
16. Timer expiry alone cannot release value or exposure.
17. Finality or independently validated no-outflow evidence is required before in-flight exposure leaves the ledger.
18. Risk can tighten immediately; relaxation requires a successor policy and controlled delay.
19. Decision records do not become execution authority.
20. Social media, screenshots, interviews, and memes remain narrative context unless separately promoted through evidence governance.

17. Limitations and nonclaims

This paper does not claim:

- that Statebook is implemented;
- exhaustive literature coverage;
- that any two listed contracts are economically or legally equivalent;
- that a candidate portfolio is executable;
- that a clearinghouse recognizes an offset;
- that an oracle, venue, bridge, benchmark, custodian, or evidence provider is correct or solvent;
- that mandatory delay would have prevented the Ostium incident;
- that the final Ostium root cause or loss is known;
- that a particular trust-to-settlement ratio is empirically calibrated;
- that the AI and macro scenarios are forecasts;
- that attestation, signatures, ZK proofs, reputation, or HSAI establish market truth, solvency, legal rights, or semantic correctness;
- custody, execution, pause, routing, signing, margin, liquidation, or settlement authority;
- production readiness, full security, benchmark evidence, SOTA, or independent reproduction.

18. Conclusion

The order book remains essential, but its unit of coherence is one identified contract. Frontier-risk markets require a higher layer because many contracts now describe related fragments of the same world while retaining different payoffs, paths, books, collateral regimes, and finality domains.

Statebook's central discipline is separation. Semantic identity is not payoff replication. Replication is not execution. Execution is not recognized capital. Atomic settlement is not legal finality. Attestation is not financial truth. Reputation is not current authorization. Delay is not security unless it covers every irreversible path and preserves the internal actions needed to remain solvent.

The viable “AWS of finance” is therefore not the venue that lists the most products. It is the substrate that lets many products compose while keeping their residuals, evidence, authority, and failure domains visible. If that substrate cannot say `Unknown`, it is not coherent. It is merely fast.

References and provenance

The complete annotated bibliography, code references, incident evidence, media classification, and source limitations are in `docs/statebook-literature-source-index.md`. The governing design boundary is `docs/integrations/statebook_terminal_payoff_and_trust_settlement.md`. Every external fact in this paper is an observation-time claim as of 15 July 2026 and may be superseded.

Appendix A — Annotated Literature, Code, Media, and Incident Index

State slice: `statebook-whitepaper-prd-and-publication-media-boundary`.

Status: `DocumentationOnly`.

Evidence ceiling: `Level0DesignNote`.

Version: 0.1, 15 July 2026.

Purpose and selection rule

This index supports the Statebook whitepaper and product requirements document. It is a bounded, decision-relevant bibliography, not a claim to enumerate every paper, market, codebase, regulation, or incident related to financial market infrastructure. Sources were selected when they materially constrain at least one of the following questions:

1. Can contracts with different product labels be represented as payoffs over common terminal states?
2. Can an economically equivalent portfolio be found and executed?
3. Can collateral and liquidation policy recognize the offset without hiding basis, oracle, liquidity, legal, or settlement risk?
4. Can externalization speed be adjusted using explicit assurance evidence, bounded loss budgets, and fail-closed state transitions?
5. Which AI, compute, energy, and macroeconomic developments are likely to create new frontier-risk underlyings or new attack capacity?

Every entry states what it supports and what it does not establish. A link, paper, repository, or incident report does not validate the proposed Statebook, authorize settlement, or raise this repository's evidence ceiling.

Observation date for mutable venue documentation, help-center pages, social posts, and repository revisions: 15 July 2026. These sources record public claims or code snapshots observed on that date; they do not establish deployment state, current availability, liquidity, incident causation, or evidence above `Level0DesignNote`.

Evidence classes

Class	Permitted use	Prohibited inference
Seminal theory	Define the intellectual lineage and formal problem	Treat assumptions as observed market facts
Peer-reviewed or working research	Support models, empirical patterns, or design alternatives	Treat one model as universal or production calibrated
Public authority or standard	Establish legal, prudential, operational, or terminology constraints	Infer legal approval for this product
Venue or vendor documentation	Describe a venue's stated product and operating model	Infer implementation correctness or solvency

Class	Permitted use	Prohibited inference
Pinned source code	Inspect a disclosed implementation path at one revision	Infer deployed bytecode identity, audit status, or exploit root cause
Incident evidence	Establish a bounded chronology or disclosed failure mode	Generalize causality beyond the evidence
Media or interview	Attribute a thesis, phrase, or operator view	Treat marketing or commentary as independent validation
Scenario source	Bound plausible macro or technology trajectories	Present a scenario as a forecast

A. Complete markets, state prices, and payoff spanning

T01 - Arrow, risk-bearing securities

- Source: Kenneth J. Arrow, “The Role of Securities in the Optimal Allocation of Risk-bearing,” Review of Economic Studies 31(2), 1964.
- Link: https://doi.org/10.2307/2296188
- Supports: state-contingent claims as a basis for allocating risk across possible future states.
- Limit: assumes a stylized economy; it does not solve contract normalization, execution, settlement, or margin recognition in fragmented venues.

T02 - Debreu, commodities indexed by state and time

- Source: Gérard Debreu, Theory of Value: An Axiomatic Analysis of Economic Equilibrium, Cowles Foundation Monograph 17, January 1959.
- Link: https://cowles.yale.edu/research/cfm-17-theory-value-axiomatic-analysis-economic-equilibrium
- Supports: treating time, location, and state as economically material parts of a commodity definition.
- Limit: provides a mathematical foundation, not an exchange architecture.

T02A - Arrow-Debreu equilibrium

- Source: Kenneth J. Arrow and Gérard Debreu, “Existence of an Equilibrium for a Competitive Economy,” Econometrica 22(3), July 1954, pp. 265-290.
- Link: https://doi.org/10.2307/1907353
- Supports: date-state commodity indexing and the equilibrium foundation for complete contingent-claim markets.
- Limit: existence under strong assumptions, not uniqueness, computability, execution, margin, or settlement.

T02B - Sequential incomplete markets

- Source: Roy Radner, “Existence of Equilibrium of Plans, Prices, and Price Expectations in a Sequence of Markets,” Econometrica 40(2), March 1972, pp. 289-303.
- Link: https://doi.org/10.2307/1909407

- Supports: sequential trading when a complete date-zero state-claim market is unavailable.
- Limit: strong theoretical assumptions and no modern venue, collateral, or settlement mechanics.

T02C - Incomplete-market welfare limit

- Source: Oliver D. Hart, “On the Optimality of Equilibrium When the Market Structure Is Incomplete,” Journal of Economic Theory 11(3), December 1975, pp. 418-443.
- Link: https://doi.org/10.1016/0022-0531(75)90028-9
- Supports: incomplete markets can produce constrained inefficiency.
- Limit: a welfare theorem, not a Statebook architecture or empirical estimate.

T02D - Generic existence in incomplete markets

- Source: Darrell Duffie and Wayne Shafer, “Equilibrium in Incomplete Markets: I: A Basic Model of Generic Existence,” Journal of Mathematical Economics 14(3), 1985, pp. 285-300.
- Link: https://doi.org/10.1016/0304-4068(85)90004-7
- Supports: generic existence of equilibrium in a specified incomplete-market model.
- Limit: an existence result under declared assumptions does not establish welfare, computational tractability, venue interoperability, or Statebook feasibility.

T03 - Ross, options and payoff completeness

- Source: Stephen A. Ross, “Options and Efficiency,” Quarterly Journal of Economics 90(1), 1976.
- Link: https://doi.org/10.2307/1886087
- Supports: options can expand the span of attainable payoffs and complete a market under stated assumptions.
- Limit: listing more options does not establish executable equivalence or collateral offsets in practice.

T04 - Breeden and Litzenberger, state-contingent prices

- Source: Douglas T. Breeden and Robert H. Litzenberger, “Prices of State-Contingent Claims Implicit in Option Prices,” Journal of Business 51(4), 1978.
- Link: https://doi.org/10.1086/296025
- Supports: option price surfaces can reveal prices for state-contingent payoffs under model assumptions.
- Limit: noisy, sparse, path-dependent, or differently settled contracts can violate the assumptions required for clean extraction.

T05 - Harrison and Pliska, no-arbitrage pricing

- Source: J. Michael Harrison and Stanley R. Pliska, “Martingales and Stochastic Integrals in the Theory of Continuous Trading,” Stochastic Processes and their Applications 11(3), 1981.
- Link: https://doi.org/10.1016/0304-4149(81)90026-0
- Supports: the no-arbitrage and equivalent-martingale-measure foundation for comparing replicating portfolios.

- Limit: real execution has discrete books, costs, funding, liquidity, credit, and legal constraints absent from the idealized model.

T05A - Option replication

- Source: Fischer Black and Myron Scholes, “The Pricing of Options and Corporate Liabilities,” Journal of Political Economy 81(3), May-June 1973.
- Link: https://doi.org/10.1086/260062
- Supports: no-arbitrage replication and decomposition of liabilities into option exposures.
- Limit: frictionless continuous-trading assumptions; no cross-venue, legal, funding, or oracle equivalence.

T05B - Compositional financial-contract semantics

- Source: Simon Peyton Jones, Jean-Marc Eber, and Julian Seward, “Composing Contracts: An Adventure in Financial Engineering,” ICFP 2000, pp. 280-292.
- Links: https://doi.org/10.1145/351240.351267 and https://www.microsoft.com/en-us/research/publication/composing-contracts-an-adventure-in-financial-engineering/
- Supports: a compositional contract DSL, denotational semantics, and executable interpretation.
- Limit: formal semantics do not establish legal enforceability, market liquidity, settlement, or correct reference data.

T05C - ACTUS contract-event semantics

- Source: ACTUS Financial Research Foundation, “Technical Specifications” and “Methodology,” living standard observed 15 July 2026.
- Links: https://www.actusfrf.org/techspecs and https://www.actusfrf.org/methodology
- Supports: mapping contract terms into deterministic contract events, states, and cash flows.
- Limit: living source requiring an implementation-version pin; not arbitrary payoff equivalence or venue authority.

T05D - Certified compilation of financial contracts

- Source: Danil Annenkov and Martin Elsman, “Certified Compilation of Financial Contracts,” PPDP 2018, pp. 5:1-5:13.
- Link: https://doi.org/10.1145/3236950.3236955
- Supports: compilation from a financial-contract language to payoff code with machine-checked semantic-preservation results under the formalized model.
- Limit: certified compilation relative to a semantics does not establish legal interpretation, reference-data correctness, liquidity, collateral recognition, or settlement authority.

T06 - Duffie, dynamic asset pricing

- Source: Darrell Duffie, Dynamic Asset Pricing Theory, 3rd ed., Princeton University Press, 2001.
- Link: https://press.princeton.edu/books/hardcover/9780691090221/dynamic-asset-pricing-theory
- Supports: formal language for dynamic trading, contingent claims, and incomplete markets.

- Limit: textbook theory does not determine Statebook product policy.

T07 - Shiller, perpetual futures

- Source: Robert J. Shiller, "Measuring Asset Values for Cash Settlement in Derivative Markets: Hedonic Repeated Measures Indices and Perpetual Futures," Journal of Finance 48(3), 1993.
- Link: https://doi.org/10.1111/j.1540-6261.1993.tb04024.x
- Supports: perpetual futures as continuing exposures tied to a reference measure rather than ordinary terminal claims.
- Limit: it does not establish that a modern crypto perpetual is equivalent to a terminal event contract or option portfolio.

T08 - Modern perpetual-contract pricing

- Source: Songrun He, Asaf Manela, Omri Ross, and Victor von Wachter, "Fundamentals of Perpetual Futures," arXiv:2212.06888, submitted 13 December 2022; version 6 revised 21 August 2024.
- Link: https://arxiv.org/abs/2212.06888
- Supports: formal treatment of funding mechanisms and prices in perpetual contracts.
- Limit: venue-specific funding, liquidation, and oracle mechanics remain decisive.

T09 - Replication of perpetual contracts

- Source: Damien Ackerer, Julien Hugonnier, and Urban Jermann, "Perpetual Futures Pricing," Mathematical Finance 36(3), July 2026, pp. 481-499; first published online 20 November 2025.
- Link: https://doi.org/10.1111/mafi.70018
- Supports: pricing and arbitrage relations for perpetual futures under explicit mechanisms.
- Limit: does not imply cross-venue execution or collateral completeness.

B. Market microstructure, information markets, and execution

T10 - Kyle, informed trading and liquidity

- Source: Albert S. Kyle, "Continuous Auctions and Insider Trading," Econometrica 53(6), 1985.
- Link: https://doi.org/10.2307/1913210
- Supports: liquidity, informed order flow, and price impact as endogenous market properties.
- Limit: a semantic payoff match does not remove market impact.

T11 - Glosten and Milgrom, bid-ask spreads

- Source: Lawrence R. Glosten and Paul R. Milgrom, “Bid, Ask and Transaction Prices in a Specialist Market with Heterogeneously Informed Traders,” Journal of Financial Economics 14(1), 1985.
- Link: https://doi.org/10.1016/0304-405X(85)90044-3
- Supports: adverse selection as a source of spreads and execution cost.
- Limit: does not cover the full operational and settlement stack.

T11A - Electronic limit-order-book structure

- Source: Lawrence R. Glosten, “Is the Electronic Open Limit Order Book Inevitable?”, Journal of Finance 49(4), September 1994, pp. 1127-1161.
- Link: https://doi.org/10.1111/j.1540-6261.1994.tb02450.x
- Supports: liquidity provision and adverse selection in a single-instrument electronic order book.
- Limit: does not establish semantic or operational equivalence across contracts.

T12 - Frequent batch auctions

- Source: Eric Budish, Peter Cramton, and John Shim, “The High-Frequency Trading Arms Race: Frequent Batch Auctions as a Market Design Response,” Quarterly Journal of Economics 130(4), 2015.
- Link: https://doi.org/10.1093/qje/qjv027
- Supports: discrete batching as a response to continuous-time speed races and mechanical arbitrage.
- Limit: batching execution is not the same as delaying asset externalization.

T13 - Prediction-market synthesis

- Source: Justin Wolfers and Eric Zitzewitz, “Prediction Markets,” Journal of Economic Perspectives 18(2), 2004.
- Link: https://doi.org/10.1257/0895330041371321
- Supports: prediction-market information aggregation, calibration, and design tradeoffs.
- Limit: binary-event calibration does not establish derivative replication or legal fungibility.

T13A - Prediction-price interpretation limit

- Source: Charles F. Manski, “Interpreting the Predictions of Prediction Markets,” Economics Letters 91(3), June 2006, pp. 425-429.
- Link: https://doi.org/10.1016/j.econlet.2006.01.004
- Supports: a binary-contract price is not automatically an objective probability outside restrictive assumptions.
- Limit: stylized risk-neutral, price-taking model.

T14 - Combinatorial information markets

- Source: Robin Hanson, “Combinatorial Information Market Design,” Information Systems Frontiers 5, 2003.
- Link: https://doi.org/10.1023/A:1022058209073
- Supports: markets over combinatorial state spaces and the computational difficulty of coherent pricing.
- Limit: computational market making does not guarantee executable portfolios across external venues.

T15 - Logarithmic market scoring rule

- Source: Robin Hanson, “Logarithmic Market Scoring Rules for Modular Combinatorial Information Aggregation,” Journal of Prediction Markets 1(1), February 2007, pp. 3-15.
- Link: https://doi.org/10.5750/jpm.v1i1.417
- Supports: bounded-loss automated market making and modular conditional markets.
- Limit: the bounded-loss property depends on the specified market-maker model, not on the Statebook as a whole.

T16 - Automated market makers and loss-versus-rebalancing

- Source: Jason Milionis, Ciamac C. Moallemi, Tim Roughgarden, and Anthony Lee Zhang, “Automated Market Making and Loss-Versus-Rebalancing,” arXiv:2208.06046, submitted 11 August 2022; version 5 revised 27 May 2024.
- Link: https://arxiv.org/abs/2208.06046
- Supports: decomposition of AMM performance and adverse selection costs.
- Limit: does not directly calibrate central-limit-order-book or clearing risk.

C. Clearing, collateral, settlement, and linked obligations

I01 - Principles for Financial Market Infrastructures

- Source: Committee on Payment and Settlement Systems and Technical Committee of IOSCO, Principles for Financial Market Infrastructures, 16 April 2012.
- Link: https://www.bis.org/cpmi/publ/d101.htm
- Supports: legal basis, governance, credit, collateral, margin, liquidity, settlement finality, operational risk, links, default management, and disclosure as distinct infrastructure obligations.
- Limit: application depends on jurisdiction, legal form, and supervisory perimeter.

I02 - Tokenisation arrangements

- Source: CPMI, Tokenisation in the Context of Money and Other Assets: Concepts and Implications for Central Banks, 2024.
- Link: https://www.bis.org/cpmi/publ/d225.htm
- Supports: atomic delivery-versus-payment, principal-risk reduction, governance, liquidity, interoperability, and netting tradeoffs.

- Limit: atomicity can reduce one risk while increasing liquidity or operational concentration; tokenisation does not itself create coherence.

I02A - Delivery versus payment

- Source: CPSS, Delivery versus Payment in Securities Settlement Systems, 9 September 1992.
- Link: https://www.bis.org/cpmi/publ/d06.htm
- Supports: principal-risk reduction and three gross/net DvP models.
- Limit: predates tokenisation; DvP linkage does not always mean technical simultaneity.

I02B - Technical and legal finality

- Source: CPMI, Distributed Ledger Technology in Payment, Clearing and Settlement: An Analytical Framework, February 2017.
- Link: https://www.bis.org/cpmi/publ/d157.htm
- Supports: separating ledger consensus or operational transfer from legally irrevocable, unconditional, and insolvency-resistant settlement finality.
- Limit: analytical framework, not governing law or a legal opinion.

I03 - Securities settlement tradeoffs

- Source: Morten Linnemann Bech, Jenny Hancock, Tara Rice, and Amber Wadsworth, “On the Future of Securities Settlement,” BIS Quarterly Review, 1 March 2020.
- Link: https://www.bis.org/publ/qtrpdf/r_qt2003i.htm
- Supports: shorter settlement and tokenisation can reduce replacement-cost exposure while increasing cash and securities liquidity needs; credit- liquidity tradeoffs remain.
- Limit: institutional analysis, not Statebook approval, legal advice, or production calibration.

I04 - Payment-versus-payment settlement

- Source: CPMI, Facilitating Increased Adoption of Payment versus Payment (PvP): Final Report, 27 March 2023.
- Link: https://www.bis.org/cpmi/publ/d216.htm
- Supports: principal risk, settlement windows, interoperability, and barriers to PvP adoption.
- Limit: Statebook portfolios can contain obligations outside eligible PvP rails.

I05 - Fast payments

- Source: CPMI, Fast Payments - Enhancing the Speed and Availability of Retail Payments, 2016.
- Link: https://www.bis.org/cpmi/publ/d154.pdf
- Supports: faster final-funds availability changes fraud, liquidity, credit, security, and consumer-protection demands.
- Limit: retail payment systems are not derivatives clearing systems.

I06 - CCP counterparty-risk netting

- Source: Darrell Duffie and Haoxiang Zhu, "Does a Central Clearing Counterparty Reduce Counterparty Risk?", Review of Asset Pricing Studies 1(1), 2011.
- Link: https://doi.org/10.1093/rapstu/rar001
- Supports: introducing a CCP can increase or decrease netting efficiency depending on product scope and existing bilateral netting sets.
- Limit: centralization is not automatically risk reducing.

I06A - Close-out netting legal basis

- Source: ISDA, 2018 Model Netting Act and Guide, 15 October 2018.
- Link: https://www.isda.org/book/2018-model-netting-act-and-guide/
- Supports: legal certainty and enforceability as prerequisites for close-out netting and capital recognition.
- Limit: model legislation only until enacted; jurisdiction-specific law and legal opinions remain necessary.

I06B - Regulated cross-margin example

- Source: CFTC, "Order Providing Exemptive Relief To Facilitate Cross-Margining of Customer Positions Cleared at Chicago Mercantile Exchange, Inc. and Fixed Income Clearing Corporation," 91 FR 20880, applicable 15 April 2026.
- Link: https://www.cftc.gov/LawRegulation/FederalRegister/final-rules/2026-07643.html
- Supports: capital offsets require defined products, accounts, margin calculations, operational coordination, customer protection, and legal conditions.
- Limit: narrow regulated arrangement, not general cross-venue offset authorization or Statebook calibration.

I07 - CCP resilience guidance

- Source: CPMI-IOSCO, Resilience of Central Counterparties: Further Guidance on the PFMI, 2017.
- Link: https://www.bis.org/cpmi/publ/d163.htm
- Supports: governance, credit and liquidity stress, margin, prefunded resources, and recovery planning.
- Limit: does not authorize a Statebook to recognize cross-venue offsets.

I08 - CME SPAN

- Source: CME Group, "SPAN Methodology Overview."
- Link: https://www.cmegroup.com/solutions/risk-management/performance-bonds-margins/span-methodology-overview.html
- Supports: scenario-based portfolio risk and performance-bond methodology as an institutional example of portfolio margin.
- Limit: methodology and parameterization are venue governed; this reference does not license SPAN or reproduce its full implementation.

I09 - OCC STANS

- Source: Options Clearing Corporation, “Margin Methodology.”
- Link: https://www.theocc.com/risk-management/margin-methodology
- Supports: Monte Carlo portfolio margin and dependence-sensitive risk measurement in a regulated clearing context.
- Limit: production methodology is not reducible to a public summary and is not a Statebook calibration source.

I10 - 24/7 derivatives risk advisory

- Source: CFTC Divisions of Clearing and Risk, Market Oversight, and Market Participants, Staff Advisory for Extending Trading and/or Clearing Operations to a 24 Hours-a-Day, 7 Days-a-Week Basis, CFTC Letter No. 26-16, 29 May 2026.
- Link: https://www.cftc.gov/csl/26-16/download
- Supports: staff expectations concerning surveillance, thin liquidity, reference-market closures, staffing, collateral, settlement, and system safeguards for registrants considering 24/7 operations.
- Limit: informational staff views; the advisory creates no new obligations and is not Commission action.

I11 - Event-contract rulemaking inquiry

- Source: CFTC, Prediction Markets, Advance Notice of Proposed Rulemaking, 91 FR 12516, published 16 March 2026.
- Link: https://www.cftc.gov/LawRegulation/FederalRegister/proposedrules/2026-05105.html
- Supports: event contracts remain subject to public-law boundaries even as payoff forms converge technologically.
- Limit: an inquiry is not a final rule or a legal opinion for this product.

I12 - Tokenized securities distinctions

- Source: U.S. Securities and Exchange Commission Divisions of Corporation Finance, Investment Management, and Trading and Markets, “Statement on Tokenized Securities,” 28 January 2026.
- Link: https://www.sec.gov/newsroom/speeches-statements/corp-fin-statement-tokenized-securities-012826-statement-tokenized-securities
- Supports: issuer-sponsored, custodial third-party, and synthetic tokenized instruments can encode materially different rights despite similar economic references.
- Limit: a joint staff statement, not a Commission rule or adjudication; its scope is U.S. federal securities law.

I12A - Benchmark governance

- Source: IOSCO Board, Principles for Financial Benchmarks, Final Report FR07/13, July 2013.
- Link: https://www.iosco.org/library/pubdocs/pdf/ioscopd415.pdf
- Supports: governance, input data, methodology, conflicts, accountability, and review for financial benchmarks.
- Limit: principles do not prove oracle integrity, manipulation resistance, or legal suitability for a specific contract.

I13 - FINOS Common Domain Model

- Source: FINOS, Common Domain Model 6.0.0, "The Common Domain Model" and "Product Model," release revision observed 15 July 2026.
- Documentation: https://cdm.finos.org/docs/common-domain-model/ and https://cdm.finos.org/docs/product-model/
- Code at the `6.0.0` tag commit `2f86259ee6002e173c7ab3e6f8b0df33ac9754cc`: https://github.com/finos/common-domain-model/tree/2f86259ee6002e173c7ab3e6f8b0df33ac9754cc
- Supports: composable payout, economic-terms, observable, product, legal- agreement, and lifecycle representations.
- Limit: representation does not supply price truth, executable equivalence, collateral recognition, legal fungibility, custody, or settlement authority.

I13A - FpML lifecycle messaging

- Source: ISDA/FpML Standards Committee, FpML 5.13 Recommendation, build 7, 12 May 2025.
- Link: https://www.fpml.org/spec/fpml-5-13-7-rec-1/
- Supports: OTC-derivative and structured-product messaging across trade lifecycle events.
- Limit: message representation is not payoff equivalence, execution, or legal fungibility.

I13B - ISO 20022 conceptual interoperability

- Source: ISO, ISO 20022-5:2026 — Financial Services — Universal Financial Industry Message Scheme — Part 5: Conceptual Interoperability and Reverse Engineering, 2nd ed., April 2026.
- Link: https://www.iso.org/standard/20022-5
- Supports: business-concept and logical-model alignment across financial messaging.
- Limit: methodology, not financial payoff normalization or settlement authority.

I13C - Unique Product Identifier

- Source: CFTC, "Order Designating the Unique Product Identifier and Product Classification System To Be Used in Recordkeeping and Swap Data Reporting," 88 FR 11790, 24 February 2023.
- Link: https://www.cftc.gov/LawRegulation/FederalRegister/final-rules/2023-03661.html
- Supports: product and reference classification and aggregation across reporting systems.
- Limit: an identifier or taxonomy is not economic equivalence.

I14 - Basel operational resilience

- Source: Basel Committee on Banking Supervision, Principles for Operational Resilience, 2021.
- Link: https://www.bis.org/bcbs/publ/d516.htm
- Supports: tolerance for disruption, dependency mapping, business continuity, and resilient operations.
- Limit: principles require institution-specific implementation.

I15 - Basel third-party and ICT risk

- Sources: Basel Committee on Banking Supervision, Principles for the Sound Management of Third-Party Risk, 10 December 2025, https://www.bis.org/bcbs/publ/d605.htm; and Information and Communication Technology Risk Management: Range of Practices, 2 June 2026, https://www.bis.org/bcbs/publ/d611.htm.
- Supports: third-party lifecycle management, supply-chain and nth-party dependencies, concentration, contingency, exit, and non-malicious ICT- incident risk in banking.
- Limit: banking-sector principles and observed practices, not a quantitative Statebook dependency model.

D. Venue convergence and programmable market infrastructure

M00 - Cloud economics behind the AWS analogy

- Sources: Amazon Web Services, "What Is Cloud Computing?", vendor documentation observed 15 July 2026, https://docs.aws.amazon.com/whitepapers/latest/aws-overview/what-is-cloud-computing.html; Michael Armbrust et al., "Above the Clouds: A Berkeley View of Cloud Computing," Technical Report UCB/EECS-2009-28, 2009, https://amplab.cs.berkeley.edu/publication/above-the-clouds-a-berkeley-view-of-cloud-computing/.
- Supports: the cloud shift from large upfront hardware ownership toward metered, on-demand service consumption and elastic capacity.
- Limit: AWS is vendor documentation, and cloud economics does not establish financial semantics, custody safety, legal finality, collateral recognition, or Statebook feasibility.

M01 - Hyperliquid portfolio margin

- Source: Hyperliquid Docs, "Portfolio Margin," alpha-labelled documentation observed 15 July 2026.
- Link: https://hyperliquid.gitbook.io/hyperliquid-docs/trading/portfolio-margin
- Supports: the page is headed "Alpha mode" while retaining several "pre-alpha" statements, and describes collectively margining selected spot balances, perpetuals, and HIP-3 DEXs within one account.
- Limit: eligibility, borrow and supply caps, fallback behavior, and mutable internally inconsistent phase labels prevent treating the page as a frozen rollout phase, broad availability, proven liquidation safety, or cross-venue capital completeness.

M02 - Hyperliquid builder-deployed perpetuals

- Source: Hyperliquid Improvement Proposal 3.
- Link: https://hyperliquid.gitbook.io/hyperliquid-docs/hyperliquid-improvement-proposals-hips/hip-3-builder-deployed-perpetuals
- Supports: permissionless builder-deployed perpetuals whose deployer controls market definition, oracle, leverage, and settlement; each DEX begins with independent books and margin, while eligible assets may enable validator-governed cross-margin.
- Limit: cross-margin is not automatic and its enabling action is documented as irreversible; mutable deployer and validator rules do not establish semantic equivalence or cross-venue capital completeness.

M03 - Kalshi BTC perpetual futures

- Source: Kalshi Help Center, “BTC Perpetual Futures — Contract Specifications,” 3 June 2026; observed 15 July 2026.
- Link: https://help.kalshi.com/en/articles/15357587-btc-perpetual-futures-contract-specifications
- Supports: product-form convergence between a regulated event-market brand and perpetual futures.
- Limit: a mutable product specification does not independently establish listing continuity, liquidity, execution quality, or portfolio offsets.

M04 - Architect compute futures

- Source: Architect Financial Technologies, “Architect Partners with Compute Index Provider Ornn to Launch Exchange-Traded Futures on GPU and RAM Prices,” 21 January 2026.
- Link: https://architect.co/insights/press/architect-ornn-compute-futures/
- Supports: announced perpetual futures linked to GPU-rental and DRAM-price indexes, pending regulatory approval.
- Additional source: Architect Financial Technologies, “Architect Financial Technologies Partners with Compute Desk to Launch Exchange-for-Physical Market for GPU Compute,” 8 July 2026, https://architect.co/insights/press/architect-compute-desk-computeconnect/.
- Limit: announcements establish product intent and intended financial-to- physical linkage, not listing approval, current availability, liquidity, benchmark quality, or hedge effectiveness.

M04A - Electricity-forward hedging

- Source: Hendrik Bessembinder and Michael L. Lemmon, “Equilibrium Pricing and Optimal Hedging in Electricity Forward Markets,” Journal of Finance 57(3), June 2002, pp. 1347-1382.
- Link: https://doi.org/10.1111/1540-6261.00463
- Supports: nonstorability, demand skewness, production risk, and hedging pressure in electricity forwards.
- Limit: GPU compute is only partially analogous to electricity and requires separate deliverability, obsolescence, benchmark, and quality modelling.

M05 - Gnosis Conditional Tokens

- Source: Gnosis, Conditional Tokens contracts and glossary.
- Link at observed commit `eeefca66`: https://github.com/gnosis/conditional-tokens-contracts/blob/eeefca66eb46c800a9aaab88db2064a99026fde5/docs/glossary.rst
- Supports: onchain condition, outcome collection, position split, merge, and redemption primitives.
- Limit: token combinatorics do not establish truthful resolution, legal equivalence, or sufficient liquidity. Revision observed 15 July 2026; source inspection is not deployed-bytecode identity, audit, liquidity, solvency, or correctness evidence.

M06 - Polymarket negative-risk adapter

- Source: Polymarket, Negative Risk Conditional Tokens Framework Adapter.
- Link at observed commit `f78b35b0`: https://github.com/Polymarket/neg-risk-ctf-adapter/blob/f78b35b0863b4308a431ca307d06f49b2ea65e78/docs/index.md
- Supports: explicit conversion among mutually exclusive event outcomes and capital-efficiency improvements within a declared condition set.
- Limit: “negative risk” conversion relies on correct market grouping and resolution; it is not general semantic equivalence. Revision observed 15 July 2026; source inspection is not deployed-bytecode identity, audit, liquidity, solvency, or correctness evidence.

M07 - Polymarket CTF exchange

- Source: Polymarket, CTF Exchange v2.
- Link at observed commit `ccc05960`: https://github.com/Polymarket/ctf-exchange-v2/tree/ccc0596074f4dfd62c944fbca4de252893b82b4b
- Supports: an open-source order and settlement surface for outcome tokens.
- Limit: revision observed 15 July 2026; source inspection is not deployed- bytecode identity, audit, liquidity, solvency, or correctness evidence.

M08 - CoW Protocol batch-auction services

- Source: CoW Protocol services.
- Link at observed commit `6b2a4ce9`: https://github.com/cowprotocol/services/tree/6b2a4ce9dcfc5731f3bfd1f68457ed0379488707
- Supports: intent-based order collection, solver competition, and batch settlement as an alternative execution architecture.
- Limit: solver-based token exchange does not solve derivative semantics or collateral recognition. Revision observed 15 July 2026; source inspection is not deployed-bytecode identity, audit, liquidity, solvency, or correctness evidence.

M09 - CoW flash-loan router

- Source: CoW Protocol flash-loan router.
- Link at observed commit `90b18bca`: https://github.com/cowprotocol/flash-loan-router/tree/90b18bca67a01dab2e55bccbc92e71f0d729b9df
- Supports: composable atomic liquidity in a bounded transaction context.
- Limit: atomic capital access can amplify failure if validation or downstream assumptions are wrong. Revision observed 15 July 2026; source inspection is not deployed-bytecode identity, audit, liquidity, solvency, or correctness evidence.

M10 - UMA Optimistic Oracle

- Source: UMA Protocol, Optimistic Oracle v3 developer quickstart.
- Link at observed commit `5c960363`: https://github.com/UMAprotocol/dev-quickstart-oov3/tree/5c9603634bdf24b6dad6bbf85195c74bde22715c
- Supports: assertion, challenge, and dispute windows as an oracle design pattern.
- Limit: optimistic resolution is not appropriate for every price, latency, or finality requirement. Revision observed 15 July 2026; source inspection is not deployed-bytecode identity, audit, liquidity, solvency, or correctness evidence.

E. Assurance, zero trust, and secure externalization

S01 - NIST Zero Trust Architecture

- Source: NIST Special Publication 800-207, Zero Trust Architecture, 2020.
- Link: https://csrc.nist.gov/pubs/sp/800/207/final
- Supports: no implicit trust based on network location or ownership and continuous, resource-specific access decisions.
- Limit: zero trust is an access architecture, not a collateral or settlement model.

S02 - IETF RATS architecture

- Source: RFC 9334, Remote ATtestation procedureS Architecture, 2023.
- Link: https://www.rfc-editor.org/rfc/rfc9334.html
- Supports: separation of evidence, appraisal policy, attestation results, and relying-party decisions.
- Limit: an attestation result establishes only the claims and trust roots actually appraised; it is not semantic correctness or financial solvency.

S03 - NIST digital identity session monitoring

- Source: NIST SP 800-63B-4, Digital Identity Guidelines: Authentication and Authenticator Management, July 2025, section 5.3, “Session Monitoring.”
- Links: https://doi.org/10.6028/NIST.SP.800-63b-4 and https://pages.nist.gov/800-63-4/sp800-63b/session/
- Supports: ongoing session risk evaluation, reauthentication, and response to changing fraud indicators.
- Limit: identity-session controls do not validate a trade or oracle price.

S04 - NIST incident-response guidance

- Source: Alexander Nelson, Sanjay Rekhi, Murugiah Souppaya, and Karen Scarfone, NIST SP 800-61 Rev. 3, Incident Response Recommendations and Considerations for Cybersecurity Risk Management: A CSF 2.0 Community Profile, April 2025.
- Link: https://doi.org/10.6028/NIST.SP.800-61r3

- Supports: preparation, detection, response, recovery, learning, and evidence preservation as an integrated risk process.
- Limit: general guidance does not determine a protocol-specific pause policy.

S05 - OpenZeppelin TimelockController

- Source: OpenZeppelin Contracts v5.6.1, `TimelockController.sol`, observed at commit `5fd1781b1454fd1ef8e722282f86f9293cacf256`.
- Link: https://github.com/OpenZeppelin/openzeppelin-contracts/blob/5fd1781b1454fd1ef8e722282f86f9293cacf256/contracts/governance/TimelockController.sol
- Supports: delayed governance execution as a transparent review and reaction window.
- Limit: a reusable control primitive is not financial-settlement proof, correct governance configuration, or evidence that every externalization path can be paused. A timelock can impede urgent response if poorly governed.

S06 - OpenZeppelin emergency-stop patterns

- Source: OpenZeppelin Contracts v5.6.1, `Pausable.sol`, observed at commit `5fd1781b1454fd1ef8e722282f86f9293cacf256`.
- Link: https://github.com/OpenZeppelin/openzeppelin-contracts/blob/5fd1781b1454fd1ef8e722282f86f9293cacf256/contracts/utils/Pausable.sol
- Supports: explicit pause states and guarded operations as reusable control patterns.
- Limit: a reusable control primitive is not financial-settlement proof, correct governance configuration, or evidence that every externalization path can be paused. Pause authority also creates governance and availability risk and cannot reverse already-final settlement.

S07 - Reported Ostium incident acknowledgement

- Source: Kyle Baird, “Ostium pauses trading after apparent $18 million vault exploit,” The Block, 15 July 2026, reporting and quoting an Ostium X post.
- Links: https://www.theblock.co/post/408450/ostium-pauses-trading-after-apparent-18-million-vault-exploit and the reported original URL https://x.com/Ostium/status/2077412452392652917.
- Supports: contemporaneous reporting quoted the project as stating that it was aware of an OLP-vault issue, had paused all trading, and was investigating.
- Limit: this is reporting of a first-party statement, not a retained first-party artifact. The original post was not independently retrievable or retained during this review. Neither source establishes final root cause, exact loss, recovery, or whether a delay would have prevented harm. The article is a developing report and may be superseded.

S08 - Reported Ostium investigation-update URL

- Source: a URL reported as an Ostium investigation update on 15 July 2026.
- Link: https://x.com/Ostium/status/2077438120354603396
- Supports: no distinct substantive claim in this index until the exact official content, timestamp, retrieval date, and content hash or archived capture are recovered.

- Limit: an inaccessible social URL cannot independently support an update claim; subsequent findings may also supersede any recovered content.

S09 - Blockaid preliminary Ostium attribution

- Source: Blockaid preliminary incident analysis, 15 July 2026.
- Link: https://x.com/blockaid_/status/2077405527428989363
- Supports: Blockaid's preliminary claim that a registered forwarder and prepared or future-dated authorized oracle reports were involved.
- Limit: no reviewed official postmortem confirms the report-preparation mechanism, signer compromise, exact loss, or final root cause. Do not paraphrase the allegation as the router accepting a request timestamp that was still in the future.

S10 - Ostium incident transaction

- Source: Arbiscan transaction `0x359f8c05b86a4409d60cfba02084334313fd94b19f74a294fb7fc4ea7d4870e0`, successful at 14:18:48 UTC on 15 July 2026.
- Link: https://arbiscan.io/tx/0x359f8c05b86a4409d60cfba02084334313fd94b19f74a294fb7fc4ea7d4870e0
- Supports: direct transaction and event-log evidence for the observed same- transaction execution sequence.
- Limit: the transaction establishes executed state transitions, not attacker identity, upstream compromise, intent, or complete root cause.

S11 - Ostium current protocol-flow documentation

- Source: Ostium, “How Ostium Works,” current mutable documentation observed 15 July 2026.
- Link: https://docs.ostium.com/protocol/how-ostium-works
- Supports: Ostium's current stated use of Chainlink and Stork oracle inputs, Chainlink and Gelato automation, immediate onchain trader-PnL payment, and daily reconciliation with an offchain hedge.
- Limit: mutable provider documentation may postdate the incident, does not prove incident-time deployed configuration, and does not independently validate provider independence or solvency claims.

S12 - Ostium withdrawal documentation

- Source: Ostium, “How to Withdraw,” current mutable documentation observed 15 July 2026.
- Links: https://docs.ostium.com/vault/getting-started/withdraw and the legacy GitBook page observed 15 July 2026, https://ostium-labs.gitbook.io/ostium-docs/vault/withdraw.
- Supports: current documentation describes a dynamic request-and-settle LP withdrawal flow, typically two to three days.
- Limit: legacy Ostium GitBook documentation described a fixed 24-48-hour flow. Neither mutable page is a frozen incident-time specification, and LP withdrawal delay does not govern immediate trader-profit payout.

S13 - Ostium pinned public close and transfer paths

- Sources at commit `8390ce497f68fb128900840e0ec30683afa945d3`:
    - https://github.com/0xOstium/smart-contracts-public/blob/8390ce497f68fb128900840e0ec30683afa945d3/src/lib/TradingCallbacksLib.sol#L620-L653
    - https://github.com/0xOstium/smart-contracts-public/blob/8390ce497f68fb128900840e0ec30683afa945d3/src/OstiumVault.sol#L731-L754
- Supports: inspection of one public source revision's profitable-close and vault-transfer paths.
- Limit: no claim is made that this commit equals incident-time deployed bytecode or contains the root cause.

S14 - Ostium pinned signer, upkeep, and timestamp paths

- Sources at commit `8390ce497f68fb128900840e0ec30683afa945d3`:
    - https://github.com/0xOstium/smart-contracts-public/blob/8390ce497f68fb128900840e0ec30683afa945d3/src/OstiumVerifier.sol#L51-L64
    - https://github.com/0xOstium/smart-contracts-public/blob/8390ce497f68fb128900840e0ec30683afa945d3/src/OstiumPrivatePriceUpKeep.sol#L77-L121
    - https://github.com/0xOstium/smart-contracts-public/blob/8390ce497f68fb128900840e0ec30683afa945d3/src/OstiumPriceRouter.sol#L70-L87
- Supports: inspection of one revision's authorization, binding, and freshness checks.
- Limit: field checks do not prove upstream data integrity or complete system safety.

S15 - Nomad bridge incident analyses

- Sources:
    - Nomad first-party RCA: https://medium.com/nomad-xyz-blog/nomad-bridge-hack-root-cause-analysis-875ad2e5aacd
    - Mandiant: https://cloud.google.com/blog/topics/threat-intelligence/dissecting-nomad-bridge-hack/
    - Coinbase: https://medium.com/the-coinbase-blog/nomad-bridge-incident-analysis-899b425b0f34
- Supports: how an initialization or validation defect can turn a bridge into a copyable, machine-speed loss path.
- Limit: the first-party RCA is supplemented by independent analysis; Nomad's bridge architecture and 2022 incident do not generalize directly to Statebook or Ostium.

S16 - Bybit 2025 incident disclosures

- Sources:
    - Bybit incident notice, 21 February 2025: https://announcements.bybit.com/en/article/incident-update---eth-cold-wallet-incident-blt292c0454d26e9140/
    - Bybit timeline, 3 March 2025: https://www.bybit.com/en/learn/this-week-in-bybit/bybit-security-incident-timeline
    - FBI attribution, 26 February 2025: https://www.ic3.gov/PSA/2025/PSA250226

- Supports: operational evidence that interface, signing, and supply-chain compromise can bypass assumptions associated with cold-wallet workflows.
- Limit: Bybit's chronology is first-party; the FBI source supports attribution and approximate loss, not a complete technical root-cause analysis. None of the sources implies that mandatory delay alone solves signing compromise.

S17 - Verichains preliminary Bybit forensic report

- Source: Thanh Nguyen / Verichains, Bybit Incident Investigation: Preliminary Report, version 1.0, 24 February 2025.
- Link: https://github.com/verichains/public-audit-reports/blob/ba7ff5154659cc0f121f59371e419f7e5d6a71e4/Bybit%20Incident%20Investigation%20-%20Preliminary%20Report%20v1.0%20(for%20public%20release).pdf
- Supports: a preliminary account of a malicious Safe UI flow and targeted signing-process compromise.
- Limit: Verichains, not Safe, authored the report; preliminary findings do not establish a universal custody-incident model.

F. AI, compute, energy, and macroeconomic context

A01 - IEA energy and AI outlook

- Source: International Energy Agency, Key Questions on Energy and AI, 16 April 2026.
- Link: https://www.iea.org/reports/key-questions-on-energy-and-ai
- Supports: the IEA central projection increases global data-centre electricity demand from about 485 TWh in 2025 to about 950 TWh in 2030, while identifying grid, chip, equipment, capital, and social-acceptance constraints.
- Limit: a conditional projection, not an observed outcome, forecast certainty, or tradable oracle.

A02 - IMF July 2026 World Economic Outlook Update

- Source: International Monetary Fund, World Economic Outlook Update: Global Economy in Crosscurrents of War and Technology, 8 July 2026.
- Link: https://www.imf.org/-/media/files/publications/weo/2026/update/july/english/text.pdf
- Supports: opposing energy/geopolitical and technology-investment forces, uneven country exposure, and a 3.0 percent 2026 and 3.4 percent 2027 global growth baseline at publication.
- Limit: the baseline can change and is not a product forecast.

A03 - World Bank June 2026 Global Economic Prospects

- Source: World Bank Prospects Group, Global Economic Prospects — June 2026, 16 June 2026.
- Link: https://thedocs.worldbank.org/en/doc/2b672b3b0415d6b66c45b66579db4ef5-0050012026/global-economic-prospects-june-2026
- Supports: a slowing global-growth baseline, energy and geopolitical stress, debt constraints, AI investment, productivity uncertainty, and uneven diffusion.
- Limit: macro projections do not validate a Statebook demand forecast.

A04 - World Bank AI foundations

- Source: World Bank, Digital Progress and Trends Report 2025: Strengthening AI Foundations, published 9 January 2026.
- Link: https://doi.org/10.1596/978-1-4648-2264-3
- Supports: cross-country differences in four AI foundations: connectivity, compute, context, and competency.
- Limit: country-level foundations do not map directly to contract liquidity.

A04A - FERC large-load integration action

- Source: Federal Energy Regulatory Commission, “FERC Launches Aggressive Targeted Action to Speed Large Load Integration,” 18 June 2026.
- Link: https://www.ferc.gov/news-events/news/ferc-launches-aggressive-targeted-action-speed-large-load-integration
- Supports: a public regulatory and grid-governance response to rapidly growing large-load interconnection demand.
- Limit: the action does not prove AI-specific market demand, forecast power prices, or establish a tradable reference methodology.

A05 - OECD AI productivity scenarios

- Source: Francesco Filippucci, Peter Gal, Katharina Laengle, and Matthias Schief, “Macroeconomic Productivity Gains from Artificial Intelligence in G7 Economies,” OECD Artificial Intelligence Papers No. 41, 30 June 2025.
- Link: https://doi.org/10.1787/a5319ab5-en
- Supports: across ten-year scenarios, estimated annual aggregate labour- productivity growth attributable to AI ranges from roughly 0.4 to 1.3 percentage points in highly exposed G7 economies, with lower estimates elsewhere.
- Limit: scenario ranges are not realized effects, global averages, or price oracles.

A05A - Firm-level generative-AI productivity evidence

- Source: Erik Brynjolfsson, Danielle Li, and Lindsey R. Raymond, “Generative AI at Work,” Quarterly Journal of Economics 140(2), 2025, pp. 889-942.
- Link: https://doi.org/10.1093/qje/qjae044
- Supports: an average productivity increase near 15 percent in one 5,172-agent customer-support setting, concentrated among less-experienced workers.
- Limit: one firm, occupation, and tool; not aggregate GDP, universal labour substitution, or a forecast.

A05B - Task-based macroeconomic AI scenario

- Source: Daron Acemoglu, “The Simple Macroeconomics of AI,” Economic Policy, 2024.
- Link: https://doi.org/10.1093/epolic/eiae042
- Supports: a task-based framework for disciplined ten-year productivity and distribution scenarios.
- Limit: outcomes are assumption-sensitive and not observed effects.

A06 - Stanford AI Index 2026

- Source: AI Index Steering Committee, Stanford Institute for Human-Centered AI, The 2026 AI Index Report, 13 April 2026.
- Links:
  - https://hai.stanford.edu/ai-index/2026-ai-index-report
  - https://hai.stanford.edu/assets/files/ai_index_report_2026_chapter_4_economy.pdf
- Supports: current indicators for investment, adoption, cost, labour, models, and economic diffusion.
- Limit: compiled indicators have differing methods and lags; they are not one causal forecast.

A07 - BIS artificial intelligence and the economy

- Source: Bank for International Settlements, Artificial Intelligence and the Economy: Implications for Central Banks, Annual Economic Report 2024, Chapter III.
- Link: https://www.bis.org/publ/arpdf/ar2024e3.htm
- Supports: productivity, inflation, labour-market, and central-bank implications of AI adoption.
- Limit: macroeconomic analysis does not determine exchange architecture.

A08 - BIS AI data governance

- Source: Juan Carlos Crisanto, Adrien Currat, Johannes Ehrentraud, and Wenguang Wu, "In Data We Trust? Emerging Policy and Supervisory Approaches to AI Data Use in Financial Services," FSI Insights No. 73, 26 March 2026.
- Link: https://www.bis.org/fsi/publ/insights73.htm
- Supports: data privacy, quality, security, third-party dependency, and market- concentration concerns in financial-services AI use.
- Limit: author views and supervisory analysis, not calibrated Statebook risk weights.

A09 - FSB AI vulnerabilities in finance

- Source: Financial Stability Board, Monitoring Adoption of Artificial Intelligence and Related Vulnerabilities in the Financial Sector, 10 October 2025.
- Link: https://www.fsb.org/2025/10/monitoring-adoption-of-artificial-intelligence-and-related-vulnerabilities-in-the-financial-sector/
- Supports: third-party concentration, market correlation, cyber risk, model risk, and governance vulnerabilities.
- Limit: monitoring categories are not calibrated Statebook risk weights.

A10 - BIS tokenisation and monetary-system trust

- Source: Bank for International Settlements, "Anchoring Trust in Money: Innovation Beyond Stablecoins," Annual Economic Report 2026, Chapter III, 23 June 2026.
- Link: https://www.bis.org/publ/arpdf/ar2026e3.htm

- Supports: trust, governance, tokenisation, composability, and monetary-system integrity as inseparable design concerns.
- Limit: the institutional model is not equivalent to permissionless venue architecture.

G. Thesis attribution and media

X01 - “AWS of finance” attribution

- Source: Ben Weiss and Leo Schwartz, “How a Harvard Grad Helped Make Hyperliquid the Biggest New Player in Crypto—with Just 11 People and No Venture Funding,” Fortune, 12 January 2026.
- Link: https://fortune.com/2026/01/12/hyperliquid-jeff-yan-defi-perpetuals-perps-exchange-defi/
- Supports: Fortune reports that Yan views Hyperliquid as the “AWS of financial infrastructure”; this supports attribution only, not independent validation.
- Limit: a founder analogy is a strategy claim, not neutral proof of convergence or infrastructure quality. Do not format the phrase as a direct Yan quotation unless a primary transcript or recording contains it.

X02 - Long-form Jeff Yan interview

- Source: VALR podcast interview with Jeff Yan, 9 July 2026.
- Link: https://blog.valr.com/blog/cedefi-valrs-integration-of-hyperliquid-ft-jeff-yan
- Supports: contemporary operator commentary on integration and market infrastructure.
- Limit: first-party promotional media, not implementation, liquidity, or benchmark evidence.

X03 - Original Statebook visual system

- Source: the SVG files under `docs/media/statebook/` created for this publication.
- Supports: original diagrams and satirical teaching artifacts that explain the architecture, completeness distinctions, and security tradeoff.
- Limit: a diagram or meme is explanatory media, not evidence.

H. Repository-grounded implementation lineage

R01 - Statebook boundary specification

- Source: `docs/integrations/statebook_terminal_payoff_and_trust_settlement.md`.
- Supports: the governing docs-only Statebook model, invariants, horizon scenarios, assurance-adjusted settlement policy, and explicit nonclaims.
- Limit: it is `Level0DesignNote` material, not runtime code or validation.

R02 - HSAI claim envelopes

- Source: `crates/hsai-claim-envelope/src/lib.rs` and its governing phase docs.
- Supports: a local implemented algebra for maturity, predicates, trust roots, provenance, conjunction, and acceptance policy.
- Limit: HSAI claim envelopes do not verify market semantics, prices, solvency, legality, or settlement.

R03 - Benchmark Semantic IR and evidence boundaries

- Source: `crates/zkbench-core/src/dsl/ir.rs`, `crates/zkbench-core/src/evidence/mod.rs`, and governing architecture docs.
- Supports: repository precedent for separating source syntax, normalized semantics, execution outcomes, Evidence Records, and Claim Boundaries.
- Limit: the benchmark IR is not a financial-contract IR and must not be reused by name alone.

R04 - HSAI admission boundary

- Source: `crates/hsai-agent-admission/src/lib.rs` at local committed revision `b4b644cd96d9b70eb21ff6681a0014245773cd0f`, inspected 15 July 2026.
- Supports: repository precedent for explicit action proposals, evidence gates, fail-closed admission, replay protection, and non-authority claims.
- Limit: the revision was not present on the configured GitHub remote when checked, the current dirty working tree is excluded, admission metadata is not settlement authority, and the current Statebook slice does not mutate this crate.

Synthesis rules used by the whitepaper and PRD

1. Semantic equivalence is narrower than shared reference. Reference, observation window, maturity, comparator, settlement source, currency, rounding, dispute, legal, and path-dependence terms all matter.
2. Semantic, payoff, execution, capital, settlement, assurance, and recovery completeness are separate verdicts. A pause or challenge window is not recovery unless all externalization paths can stop, in-flight state can reconcile, evidence survives, and reopening cannot duplicate or orphan liabilities.
3. A perpetual is a continuing hedge profile, not silently a terminal claim.
4. Atomic settlement and delayed externalization solve different problems. Atomic linked exchange can reduce principal risk; a challenge window can reduce irreversible propagation after suspicious state transitions.
5. Assurance determines release permission and exposure, not truth. Hard gates decide whether release is possible, loss budgets decide how much, delay tiers decide when, and a challenge queue decides what happens next.
6. No evidence class substitutes for another. A signature is not semantic correctness; source code is not deployed-state proof; documentation is not solvency; a scenario is not a forecast; a meme is not evidence.
7. All external facts are observation-time facts. Venue designs, regulations, incidents, and macro baselines can change after 15 July 2026.

Nonclaims

This index does not claim exhaustive literature coverage, legal advice, investment advice, independent incident forensics, empirical calibration, production readiness, semantic correctness, full security, benchmark evidence, or authorization to implement exchange, custody, pause, routing, oracle, margin, liquidation, signing, or settlement powers.